Quantitative Convergence Analysis of Stochastic Maximum Principles for Mean-Field Games with Common Noise

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Abstract

This thesis develops a mathematically rigorous and computationally implementable framework for mean-field games with common noise. The central object of study is the *stochastic mean-field equilibrium*: the Nash equilibrium of a large population of interacting agents subject to both idiosyncratic and common sources of randomness, characterised through a system of forward-backward stochastic differential equations (FBSDEs) coupled with a stochastic Fokker–Planck equation. The Wasserstein distance $W_2(\mu_t, \mu^*)$ between the evolving empirical measure and the equilibrium measure serves throughout as the primary metric for quantifying convergence and stability.

The thesis establishes five principal contributions:

1. **Quantitative well-posedness** (Chapter 3, Theorem 3.1). Under standard regularity assumptions on the Hamiltonian and coefficients, the FBSDE characterisation of the common-noise MFG equilibrium admits a unique solution, with an explicit parameter-dependent contraction constant $\rho < 1$ in a weighted Wasserstein metric. The contraction bound is sharp enough to serve as a computable design criterion for numerical algorithms.

2. **Numerical convergence** (Chapter 4, Theorem 4.1). A combined Euler–Maruyama/particle/SPD scheme for the coupled FBSDE–Fokker–Planck system converges with weak Wasserstein error $O(\Delta t^{1/2} + \Delta x + M^{-1/2})$, where Δt is the time step, Δx the spatial mesh, and M the particle count. The scheme satisfies a CFL-type stability condition that couples the temporal and spatial resolutions. Convergence is verified numerically against the closed-form linear-quadratic solution, with a corrected Riccati system (see below) used as the benchmark.

3. **Particle methods and propagation of chaos** (Chapter 5). An interacting particle system of N agents approximates the McKean–Vlasov dynamics with strong error $O(N^{-1/2})$ (Fournier–Guillin rate), uniformly in time under dissipation conditions. Variance-reduction techniques reduce this to $O(N^{-1})$ for smooth test functions. The particle scheme parallelises naturally; complexity and GPU implementation strategies are analysed.

4. **Corrected linear-quadratic master equation** (Appendix A). The canonical LQ-MFG master equation is re-derived from first principles, identifying and correcting sign errors in the A_t Riccati coefficient and missing Lions-derivative coupling terms in the B_t–F_t equations that appeared in prior derivations. The corrected system is verified by three independent methods: symbolic PDE-residual substitution, two independent numerical ODE solvers (LSODA and Radau), and structural consistency with the two-state LQR matrix Riccati equation.

5. **Lean 4 formal verification** (Chapter 6). Key mathematical properties are formalised in Lean 4 targeting Mathlib, including Gronwall's inequality, optimality of the linear feedback control in the LQ-MFG, the contraction factor in weighted Wasserstein metric, and the closed-form Wasserstein-2 distance between Gaussian measures. The proofs follow standard Mathlib idioms; compilation against a live Mathlib installation is handled by the repository's continuous-integration workflow.

The honest limitations of the present work are documented throughout. The subsequential (rather than full) nature of the actor-critic convergence theory, the restriction to the LQ special case for explicit numerical illustration, and the gap in the matrix Riccati re-derivation for the general multi-dimensional model are each flagged where they appear. The Lean 4 proofs are reviewed for mathematical correctness but should be rebuilt via `lake build` before being relied on as machine-verified, as Mathlib's API evolves.

The work establishes a foundation for several natural extensions: jump-diffusion common noise, mean-field control (social optimum), deep Galerkin and physics-informed neural network solvers, and the Latała–Oleszkiewicz inequality as a route from subsequential to full convergence. These are discussed in Chapter 7.

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Chapter 1

Introduction

1.1 Overview

This thesis studies *mean-field games with common noise*: stochastic differential games in which a continuum of agents interact through a shared empirical distribution, and where all agents are simultaneously exposed to a common Brownian motion in addition to their individual idiosyncratic noise. The central mathematical objects are:

- the *conditional measure flow* $t \mapsto \mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$, the conditional distribution of the representative agent's state given the common-noise filtration;
- the *master equation*, a nonlinear second-order PDE on the space of probability measures $\mathcal{P}_2(\mathbb{R}^d)$ governing the value function $U(t, x, \mu)$ of the game; and
- the *FBSDE characterisation*, a system of coupled forward-backward stochastic differential equations (FBSDEs) whose solution characterises the Nash equilibrium through the Stochastic Maximum Principle.

The thesis makes four contributions: a quantitative contraction theorem for the FBSDE fixed-point map (Chapter 3); a convergent and stable numerical scheme for the coupled FBSDE–Fokker–Planck system (Chapter 4); a rigorous particle-method analysis including propagation-of-chaos estimates and computational complexity (Chapter 5); and a partial formal verification of the core estimates in Lean 4 (Chapter 6). Throughout, the canonical linear-quadratic MFG serves as the benchmark, with a corrected master-equation derivation (Appendix A) providing the closed-form reference solution.

1.2 What Is a Mean-Field Game?

1.2.1 The N-Player Stochastic Differential Game

Consider N agents indexed $i = 1, \ldots, N$, each with state process $X_t^i \in \mathbb{R}^d$ satisfying

$$dX_t^i = b(t, X_t^i, \mu_t^N, \alpha_t^i)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0, \qquad X_0^i \sim \mu_0, \tag{1.1}$$

where $W^1, \ldots, W^N$ are independent Brownian motions (idiosyncratic noise), W^0 is a common Brownian motion shared by all agents, and $\mu_t^N = \frac{1}{N}\sum_{j=1}^N \delta_{X_t^j}$ is the empirical measure. Each agent i chooses a control α_t^i to minimise its individual cost

$$J^i(\alpha^i; \alpha^{-i}) = \mathbb{E}\left[\int_0^T f(t, X_t^i, \mu_t^N, \alpha_t^i)\, dt + g(X_T^i, \mu_T^N)\right].$$

The Nash equilibrium $(\alpha^{1,*}, \dots, \alpha^{N,*})$ satisfies $J^i(\alpha^{i,*}; \alpha^{-i,*}) \le J^i(\alpha^i; \alpha^{-i,*})$ for all admissible α^i and all i.

1.2.2 The Mean-Field Limit

As $N \to \infty$, under regularity and monotonicity conditions on b, f, g, the empirical interaction $\mu_t^N \to \mu_t$ in a suitable sense, and the optimal control $\alpha^{i,*} \to \alpha^*(t, X_t^i, \mu_t)$ for a *common* feedback function α^*. The limit is described by a McKean–Vlasov SDE:

$$dX_t = b(t, X_t, \mu_t, \alpha^*(t, X_t, \mu_t))\, dt + \sigma\, dW_t + \sigma_0\, dW_t^0, \qquad X_0 \sim \mu_0, \tag{1.2}$$

where $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$.

Key structural observation. In the presence of common noise, the conditional measure μ_t is itself a stochastic process adapted to the common-noise filtration $\mathcal{F}_t^{W^0}$. Individual states X_t^i are adapted to the enlarged filtration generated by both W^0 and W^i, but the equilibrium object μ_t is measurable with respect to W^0 alone. This asymmetry distinguishes the common-noise theory from the classical (idiosyncratic-only) MFG, where μ_t is deterministic.

1.2.3 The ε-Nash Property

The mean-field equilibrium is justified by the following approximation result, established rigorously in Chapter 3:

Theorem 1.1 (ε-Nash Property, informal)**.** *Let α^* be the mean-field equilibrium control. For any $\varepsilon > 0$, there exists N_0 such that for all $N \ge N_0$:*

$$J^i(\alpha^*; \alpha^{*,-i}) \le \inf_{\alpha^i} J^i(\alpha^i; \alpha^{*,-i}) + \varepsilon,$$

where $\alpha^{*,-i}$ means all agents except i use the mean-field control.

This justifies solving the computationally tractable mean-field problem in place of the intractable N-player game.

1.3 The Common-Noise FBSDE Characterisation

The mean-field equilibrium is characterised via the Stochastic Maximum Principle. Let $H(t, x, \mu, p, \alpha) = p \cdot b(t, x, \mu, \alpha) - f(t, x, \mu, \alpha)$ be the Hamiltonian and $H^*(t, x, \mu, p) = \sup_\alpha H(t, x, \mu, p, \alpha)$ the maximised Hamiltonian. The equilibrium $(X^*, \mu^*, \alpha^*, p, q, r)$ satisfies:

$$dX_t^* = b(t, X_t^*, \mu_t^*, \alpha_t^*)\, dt + \sigma\, dW_t + \sigma_0\, dW_t^0, \qquad X_0^* \sim \mu_0, \tag{1.3}$$

$$dp_t = -H_x(t, X_t^*, \mu_t^*, p_t)\, dt + q_t\, dW_t + r_t\, dW_t^0, \qquad p_T = -g_x(X_T^*, \mu_T^*), \tag{1.4}$$

$$\alpha_t^* = \arg\max_\alpha H(t, X_t^*, \mu_t^*, p_t, \alpha), \tag{1.5}$$

$$\mu_t^* = \mathcal{L}(X_t^* \mid \mathcal{F}_t^{W^0}). \tag{1.6}$$

The four equations (1.3)–(1.6) form the FBSDE system studied in Chapter 3. The forward equation (1.3) is driven by the common noise; the backward equation (1.4) is a *stochastic* BSDE because μ_t^* is stochastic. The consistency condition (1.6) couples the forward and backward equations through the distribution of the state.

1.4 The Master Equation

The master equation is the fundamental PDE governing the value function $U : [0,T] \times \mathbb{R}^d \times \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ of the mean-field game. It reads

$$-\partial_t U = H^*(t,x,\mu,D_x U) + \bar{b}(t,\mu) \cdot D_\mu U + \frac{\sigma^2 + \sigma_0^2}{2} \Delta_x U + \sigma_0^2 \operatorname{tr}(D_x D_\mu U) + \frac{\sigma_0^2}{2} \int D_\mu^2 U(\tilde{x},\tilde{x})\, \mu(d\tilde{x}), \tag{1.7}$$

with terminal condition $U(T,x,\mu) = g(x,\mu)$, where $D_\mu U$ denotes the Lions derivative of U with respect to μ, and $\bar{b}(t,\mu)$ is the aggregate drift of the equilibrium measure flow. The terms in (1.7) have the following roles:

- $H^* + \bar{b} \cdot D_\mu U$: individual optimisation and mean-field coupling;
- $\frac{\sigma^2+\sigma_0^2}{2} \Delta_x U$: smoothing from the total noise;
- $\sigma_0^2 \operatorname{tr}(D_x D_\mu U) + \frac{\sigma_0^2}{2} \int D_\mu^2 U \, d\mu$: Itô correction terms from the stochasticity of μ_t itself, present only when $\sigma_0 > 0$.

The common-noise terms (last two) are absent from the classical deterministic-measure MFG master equation and constitute the primary analytic challenge of this work.

1.5 State of the Art

1.5.1 Classical Mean-Field Games

Mean-field games were introduced independently by Lasry and Lions (Lasry and Lions, 2007) and by Huang, Caines, and Malhamé (?). Both papers established the fundamental $N \to \infty$ limit, the PDE system (HJB coupled with Fokker–Planck), and the ε-Nash property. Cardaliaguet (?) gave a definitive PDE treatment of the master equation for the deterministic-measure case. The probabilistic theory was developed comprehensively by Carmona and Delarue (Carmona and Delarue, 2018), establishing well-posedness under monotonicity conditions via FBSDEs.

1.5.2 Common-Noise Mean-Field Games

The extension to common noise was initiated by Carmona, Delarue, and Lachapelle (?), who introduced the conditional measure flow $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$ as the correct state variable. Ahuja, Ren, and Yang (Ahuja et al., 2019) established well-posedness under displacement monotonicity without the strong convexity assumptions of earlier work. The master equation (1.7) with common-noise terms was analysed by Cardaliaguet, Delarue, Lasry, and Lions (Cardaliaguet et al., 2019). SPDE regularity for common-noise MFGs was established by Cardaliaguet, Seeger, and Souganidis (Cardaliaguet et al., 2025).

1.5.3 Numerical Methods

Numerical methods for MFGs (without common noise) include the finite-difference schemes of Achdou and Capuzzo-Dolcetta (?), the monotone schemes of Carlini and Silva, and the machine-learning approaches of Carmona and Laurière (?). For common-noise MFGs, the particle approximation approach is natural (the empirical measure approximates μ_t^N directly), but convergence proofs require control of the interaction between the particle system and the common noise. Chapter 4 develops a convergent combined scheme and Chapter 5 analyses the particle approximation theory in depth.

1.5.4 Propagation of Chaos

The classical propagation-of-chaos theory is due to McKean (?) and Sznitman (?). For McKean–Vlasov SDEs, quantitative convergence rates in Wasserstein distance were established by Fournier and Guillin (Fournier and Guillin, 2015). The common-noise setting requires conditional chaos estimates – bounding $W_2(\mu_t^N, \mu_t)$ given W^0 – which are developed in Chapter 5.

1.6 The Three Research Questions

This thesis addresses three interconnected questions:

Q1 (Well-posedness). Under what conditions on the Hamiltonian, the coupling through the distribution, and the noise coefficients does the common-noise MFG equilibrium exist uniquely? What are the explicit parameter-dependent contraction constants, and how do they depend on σ_0?

Q2 (Numerics). What is the optimal convergence rate of numerical schemes for the common-noise FBSDE–Fokker–Planck system? How does the presence of common noise affect the CFL stability condition, and what is the computational complexity?

Q3 (Particle approximation). How rapidly does the N-particle system converge to the mean-field limit, uniformly in time? Can variance-reduction techniques improve the $O(N^{-1/2})$ Fournier–Guillin rate for smooth test functions? What is the computational cost of a parallel particle implementation?

1.7 Principal Contributions

The principal mathematical contributions of this thesis are:

Theorem 1.2 (Quantitative Well-posedness, Chapter 3). *Under Assumptions A1–A5 (Lipschitz, monotone, bounded coefficients), the fixed-point map $\Phi : \mathcal{C}([0,T]; \mathcal{P}_2(\mathbb{R}^d)) \to \mathcal{C}([0,T]; \mathcal{P}_2(\mathbb{R}^d))$ characterising the common-noise MFG equilibrium is a contraction in the weighted Wasserstein distance $d_\beta(\mu, \nu) = \sup_t e^{-\beta t} W_2(\mu_t, \nu_t)$ with explicit contraction constant*

$$\rho = \frac{C_\alpha^2 \cdot C_{\mathrm{BSDE}}}{2\beta} < 1,$$

for β large enough relative to the Lipschitz constants. Consequently, the equilibrium $(X^*, \mu^*, \alpha^*, p, q, r)$ exists and is unique.

Theorem 1.3 (Numerical Convergence, Chapter 4). *The combined Euler–Maruyama / particle / finite-difference scheme for* (1.3)–(1.6) *converges with weak error*

$$\mathbb{E}[W_2(\mu_T^{\Delta}, \mu_T^*)] \le C\left(\Delta t^{1/2} + \Delta x + M^{-1/2}\right),$$

where C depends only on the model parameters and horizon T, under the CFL condition $\Delta t = O(\Delta x^2)$.

Theorem 1.4 (Propagation of Chaos, Chapter 5). *Let $(\mu_t^N)_{t\in[0,T]}$ be the empirical measure of the N-particle interacting system. Under dissipativity Assumption D1,*

$$\sup_{t\in[0,T]} \mathbb{E}\left[W_2(\mu_t^N, \mu_t^*)^2 \mid W^0\right] \le \frac{C}{N},$$

uniformly over common-noise realisations W^0, where C depends on the coefficients, T, d, and the initial variance $\mathbb{E}[|X_0|^2]$ but not on N.

Proposition 1.5 (Corrected LQ Master Equation, Appendix A). *The linear-quadratic common-noise MFG (Example 1.6) admits the explicit value function $V(t,x,\mu) = -\frac{1}{2}A_t x^2 - B_t x\bar{\mu} - \frac{1}{2}C_t\bar{\mu}^2 - D_t x - E_t\bar{\mu} - F_t$ where $A_t, \dots, F_t$ satisfy the corrected Riccati system derived in Appendix A. The corrected system differs from earlier derivations in (i) the signs of the A_t^2/λ and c_1 terms in $\dot{A}_t$, and (ii) the Lions-derivative coupling terms in $\dot{B}_t$–$\dot{F}_t$.*

1.8 The Canonical Linear-Quadratic Benchmark

Throughout the thesis, the following linear-quadratic model serves as the canonical benchmark against which numerical schemes are validated.

Example 1.6 (LQ-MFG, canonical benchmark).

$$\begin{aligned} \text{State:} \quad & dX_t = \left[\kappa(\bar{\mu}_t - X_t) + \alpha_t\right]dt + \sigma\, dW_t + \sigma_0\, dW_t^0, \\ \text{Running cost:} \quad & f(t,x,\mu,\alpha) = \tfrac{c_1}{2}x^2 + \tfrac{\lambda}{2}\alpha^2, \\ \text{Terminal cost:} \quad & g(x,\mu) = \tfrac{c_2}{2}(x-\bar{\mu})^2, \end{aligned}$$

where $\bar{\mu}_t = \int x\, \mu_t(dx)$ is the population mean, $\kappa > 0$ is the mean-reversion rate, $\lambda > 0$ is the control cost, and $c_1, c_2 > 0$ are running and terminal state costs. Baseline parameters: $\kappa = 0.5$, $\lambda = 1$, $c_1 = 2$, $c_2 = 5$, $\sigma = 0.2$, $\sigma_0 = 0.3$, $T = 1$.

This model arises in many applications – coordination games, interacting oscillators, collective motion – and is one of the few MFG models with a fully explicit equilibrium. It satisfies all standing assumptions (A1–A5, D1) and thus serves as a rigorous benchmark for both the theoretical estimates and the numerical schemes.

1.9 Thesis Roadmap

Chapter 2 reviews the mathematical prerequisites: Wasserstein geometry, Itô calculus with common noise, Sobolev spaces and SPDEs, Lions derivatives, and the Stochastic Maximum Principle.

Chapter 3 establishes the quantitative well-posedness of the common-noise MFG (Theorem 1.2), including the contraction estimate and the ε-Nash property.

Chapter 4 develops numerical approximation: discretisation of the FBSDE system, particle approximation of the measure flow, finite-difference approximation of the Fokker–Planck SPDE, combined error estimates, and CFL analysis. All schemes are validated against the LQ benchmark.

Chapter 5 analyses particle methods in depth: the N-particle McKean–Vlasov system, conditional propagation of chaos, empirical measure convergence rates, variance-reduction techniques, and parallel / GPU implementation.

Chapter 6 presents the Lean 4 formal verification of key estimates (Gronwall, contraction factor, LQ optimality, Wasserstein-2 Gaussian formula).

Chapter 7 discusses future directions: jump-diffusion extensions, the social optimum (mean-field control), deep Galerkin solvers, and the route from subsequential to full convergence via the Latała–Oleszkiewicz inequality.

Appendix A provides the full derivation and correction of the LQ master equation, verified three independent ways.

1.10 Key Notation and Standing Assumptions

Notation. $\mathcal{P}_2(\mathbb{R}^d)$ denotes the space of Borel probability measures on $\mathbb{R}^d$ with finite second moment. $W_2(\mu,\nu)$ is the Wasserstein-2 distance. $\bar{\mu} = \int x\,\mu(dx)$ is the mean. $C([0,T];\mathcal{P}_2)$ is the space of continuous measure-valued paths. All filtrations are assumed complete. $\mathbb{E}_0[\cdot] = \mathbb{E}[\cdot \mid \mathcal{F}_T^{W^0}]$ denotes conditional expectation given the full common-noise path.

Assumptions A1–A5.

- A1. *Lipschitz.* b, f, g are jointly Lipschitz in (x,μ) uniformly in t, with Lipschitz constant L.
- A2. *Monotonicity.* $\langle b(t,x,\mu,\alpha) - b(t,x',\mu,\alpha), x - x'\rangle \le -\kappa_0|x-x'|^2$ for some $\kappa_0 > 0$ (drift dissipativity).
- A3. *Uniform concavity.* $H(t,x,\mu,p,\cdot)$ is uniformly λ_0-concave in α, with $\lambda_0 > 0$.
- A4. *Monotone coupling.* $\mu \mapsto f(t,x,\mu,\alpha)$ and $\mu \mapsto g(x,\mu)$ satisfy the Lasry–Lions monotonicity condition.
- A5. *Bounded volatility.* $\sigma, \sigma_0 > 0$ are constants.

Assumption D1 (Dissipativity for chaos). $\langle b(t,x,\mu,\alpha^*(t,x,\mu)) - b(t,x',\mu,\alpha^*(t,x',\mu)), x - x'\rangle \le -\kappa_1|x-x'|^2$ for some $\kappa_1 > 0$, uniformly in t, μ.

Chapter 2

Mathematical Preliminaries

2.1 Overview

This chapter develops the complete mathematical toolkit required for all subsequent chapters. No prior knowledge of optimal transport or stochastic partial differential equations is assumed; every result is stated with either a complete proof or a detailed proof sketch, with references to the authoritative sources for full details. The standing assumptions A1–A7 referenced throughout are collected in Section 1.9 of Chapter 1.

The chapter covers six interlocking topics:

1. **Wasserstein-2 space** $(\mathcal{P}_2(\mathbb{R}^d), W_2)$: optimal transport, metric space structure, geodesics, weak convergence, rates of convergence for empirical measures, and the Lions derivative.

2. **Brownian motion and Itô calculus**: construction, quadratic variation, the Itô integral, Itô's formula, and stochastic differential equations.

3. **Fokker–Planck SPDE**: derivation from Itô's formula, weak formulation, strong form, Sobolev regularity, and the Gelfand triple framework. We clarify the distinction between the linear Fokker–Planck SPDE and the nonlinear Kushner–Zakai equation.

4. **Martingale theory**: martingales, stopping times, Doob's inequality, Burkholder–Davis–Gundy, martingale representation, and Girsanov's theorem.

5. **Sobolev spaces and functional analysis**: Sobolev embedding, Hahn–Banach extension, Lumer–Phillips theorem, spectral gap, and log-Sobolev inequalities.

6. **Stochastic Maximum Principle**: the control problem, Hamiltonian, adjoint BSDE, optimality conditions, and connections to Hamilton–Jacobi–Bellman equations.

A summary table at the end of the chapter (Section 2.8) maps each tool to its first use in the thesis, serving as a quick reference. For readers seeking a deeper treatment, we recommend Villani (2009) for optimal transport, Carmona–Delarue (2018, Vol. I, Chapters 1–2) for Wasserstein analysis in the MFG context, and Karatzas–Shreve (1991) for stochastic calculus.

2.2 Wasserstein-2 Space

2.2.1 Probability Measures with Finite Second Moment

Let $\mathcal{B}(\mathbb{R}^d)$ denote the Borel σ-algebra on $\mathbb{R}^d$. The space $\mathcal{P}_2(\mathbb{R}^d)$ consists of all Borel probability measures μ on $\mathbb{R}^d$ with finite second moment:

$$\mathcal{P}_2(\mathbb{R}^d) := \left\{ \mu \in \mathcal{P}(\mathbb{R}^d) \;\middle|\; \int_{\mathbb{R}^d} |x|^2 \, \mu(dx) < \infty \right\}. \tag{2.1}$$

Elements of $\mathcal{P}_2(\mathbb{R}^d)$ arise naturally in our setting:

- As the law $\mu = \mathcal{L}(X)$ of an $\mathbb{R}^d$-valued random variable with $\mathbb{E}[|X|^2] < \infty$.
- As the empirical measure $\mu^N = \frac{1}{N}\sum_{i=1}^N \delta_{x_i}$ when the particles satisfy $\frac{1}{N}\sum_i |x_i|^2 < \infty$.
- As the mean-field distribution μ_t in the MFG system, provided the agents' positions remain square-integrable in expectation.

2.2.2 Optimal Transport and the Wasserstein Distance

Definition 2.1 (Coupling and Wasserstein-2 Distance)**.** A coupling of $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ is a probability measure $\pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ with marginals μ and ν: $\pi(\cdot \times \mathbb{R}^d) = \mu$, $\pi(\mathbb{R}^d \times \cdot) = \nu$. The set of all couplings is denoted $\Pi(\mu, \nu)$. The Wasserstein-2 distance between μ and ν is

$$W_2(\mu, \nu) := \left(\inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} |x - y|^2 \, \pi(dx, dy) \right)^{1/2}. \tag{2.2}$$

The infimum in (2.2) is attained: there exists an optimal coupling $\pi^* \in \Pi(\mu, \nu)$ achieving the infimum (Villani, 2009, Theorem 5.10). The optimal π^* is supported on the subdifferential of a convex function, the Brenier potential.

Theorem 2.2 (Brenier's Theorem, 1991)**.** *Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ with μ absolutely continuous with respect to Lebesgue measure. Then there exists a unique (up to μ-a.e. equality) convex function $\varphi : \mathbb{R}^d \to \mathbb{R}$ such that the map $T = \nabla\varphi$ satisfies $T_\# \mu = \nu$ (i.e., $\nu(A) = \mu(T^{-1}(A))$ for all A) and*

$$W_2(\mu, \nu)^2 = \int_{\mathbb{R}^d} |x - \nabla\varphi(x)|^2 \, \mu(dx).$$

The optimal coupling is $\pi^* = (\mathrm{id} \times T)_\# \mu$, i.e., the law of $(X, T(X))$ when $X \sim \mu$.

Proof sketch. The proof proceeds in three steps: (i) Kantorovich duality reduces the problem to finding a pair of conjugate convex functions (φ, φ^*); (ii) the optimality condition $y \in \partial\varphi(x)$ for π^*-a.e. (x, y) implies the transport is a gradient map; (iii) uniqueness follows from strict convexity of the cost $c(x, y) = |x - y|^2$. See Villani (2009, Ch. 9) for the complete argument. □

Remark 2.3 (Discrete measures and regularisation). Brenier's Theorem requires absolute continuity of the source measure μ. In particle-method applications (Chapter 5), the empirical measure μ_t^N is a sum of Dirac masses, and the equilibrium measure μ_t^* is approximated from discrete particle positions. The optimal transport between discrete measures is not uniquely given by a gradient of a convex function in the classical sense. The Sinkhorn algorithm (Section 2.2.3) bridges this gap: by adding an entropic penalty, the discrete measures are effectively "blurred" into continuous approximations, recovering a gradient-like structure and enabling stable computation.

Example 2.4 (One-dimensional case: sorted coupling)**.** In $d = 1$, the optimal transport has a particularly elegant form. For $\mu, \nu \in \mathcal{P}_2(\mathbb{R})$ with distribution functions F_μ, F_ν and quantile functions F_μ^{-1}, F_ν^{-1}:

$$W_2(\mu, \nu)^2 = \int_0^1 |F_\mu^{-1}(u) - F_\nu^{-1}(u)|^2 \, du. \tag{2.3}$$

The optimal map is $T(x) = F_\nu^{-1}(F_\mu(x))$ — transport the u-quantile of μ to the u-quantile of ν (sorted coupling). This is computed exactly in $O(N \log N)$ by sorting the particles. **The synchronous coupling used in Chapter 3 for common-noise cancellation can be explicitly constructed in 1-D via this quantile coupling.**

2.2.3 The Sinkhorn Algorithm

For computational purposes, we regularise the Kantorovich problem with an entropic penalty:

$$W_{2,\epsilon}(\mu, \nu)^2 := \inf_{\pi \in \Pi(\mu,\nu)} \int |x - y|^2 \pi + \epsilon \, \mathrm{KL}(\pi \| \mu \otimes \nu), \tag{2.4}$$

where $\mathrm{KL}(\pi \| \mu \otimes \nu) = \int \log \frac{d\pi}{d(\mu \otimes \nu)} \, d\pi$. The solution has the form

$$\pi_\epsilon^*(dx, dy) = u(x) K(x, y) v(y) \, \mu(dx) \nu(dy),$$

where $K(x, y) = e^{-|x-y|^2/\epsilon}$ is the Gaussian kernel, and (u, v) satisfies the Sinkhorn fixed-point equations:

$$u_i = a_i \Big/ \sum_j K_{ij} v_j, \qquad v_j = b_j \Big/ \sum_i K_{ij} u_i, \tag{2.5}$$

where $a_i = \mu(\{x_i\})$, $b_j = \nu(\{y_j\})$. This converges in $O(N^2/\epsilon^2)$ operations (Cuturi, 2013). In Chapter 4, this regularised distance will be used to compute Wasserstein gradients efficiently.

2.2.4 Metric Space Structure

Theorem 2.5 (($\mathcal{P}_2(\mathbb{R}^d), W_2$) is a Complete Separable Metric Space)**.** *The Wasserstein-2 distance satisfies:*

(i) Non-degeneracy: $W_2(\mu, \nu) = 0 \iff \mu = \nu$*;*

(ii) Symmetry: $W_2(\mu, \nu) = W_2(\nu, \mu)$*;*

(iii) Triangle inequality: $W_2(\mu, \rho) \leq W_2(\mu, \nu) + W_2(\nu, \rho)$*;*

(iv) Completeness: every Cauchy sequence in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ *converges to an element of* $\mathcal{P}_2(\mathbb{R}^d)$*;*

(v) Separability: $\mathcal{P}_2(\mathbb{R}^d)$ contains a countable dense subset.

Proof of (iii) via gluing lemma. Given μ, ν, ρ, let $\pi_{12} \in \Pi(\mu, \nu)$ and $\pi_{23} \in \Pi(\nu, \rho)$ be any couplings. By the gluing lemma (Villani, 2009, Lemma 7.6), there exists $\pi_{123} \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R}^d)$ with marginals (π_{12}, π_{23}), i.e., the joint distribution of (X, Y, Z) where $(X, Y) \sim \pi_{12}$, $(Y, Z) \sim \pi_{23}$. The marginal $\pi_{13} = (\mathrm{proj}_{13})_{\#}\pi_{123}$ is a coupling of μ and ρ. By Minkowski's inequality:

$$W_2(\mu, \rho) \leq \left(\int |x - z|^2 \pi_{13}(dx, dz) \right)^{1/2} = \mathbb{E}[|X - Z|^2]^{1/2} \leq \mathbb{E}[|X - Y|^2]^{1/2} + \mathbb{E}[|Y - Z|^2]^{1/2}.$$

Optimising over π_{12} and π_{23} gives the triangle inequality. The remaining properties are proved similarly. □

2.2.5 Weak Convergence in $\mathcal{P}_2$

Theorem 2.6 (Characterisation of W_2-convergence)**.** *The following are equivalent for $\mu_n, \mu \in \mathcal{P}_2(\mathbb{R}^d)$:*

(i) $W_2(\mu_n, \mu) \to 0$;

(ii) $\mu_n \xrightarrow{w} \mu$ weakly and $\int |x|^2 \, \mu_n(dx) \to \int |x|^2 \, \mu(dx)$;

(iii) For every continuous $f : \mathbb{R}^d \to \mathbb{R}$ with $|f(x)| \leq C(1 + |x|^2)$: $\int f \, d\mu_n \to \int f \, d\mu$.

Condition (ii) shows that W_2-convergence is strictly stronger than weak convergence: it additionally requires convergence of the second moment. **This is precisely the correct topology for the MFG analysis, since $\mu_t = \mathcal{L}(X_t)$ and $\mathbb{E}[|X_t|^2]$ must remain finite and continuous in t.**

2.2.6 Geodesics and the Lions Derivative

The space $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ has a rich geometric structure that motivates the Lions derivative used in the master equation.

Definition 2.7 (Wasserstein Geodesic)**.** The geodesic (shortest path) from μ_0 to μ_1 in $(\mathcal{P}_2(\mathbb{R}^d), W_2)$ is the curve

$$\mu_t := \big((1 - t)\,\mathrm{id} + tT\big)_{\#}\mu_0, \qquad t \in [0, 1],$$

where $T = \nabla\varphi$ is the Brenier optimal transport map from μ_0 to μ_1. This interpolates linearly in position space: the x-atom of μ_0 travels in a straight line to $T(x)$ at constant speed.

Proposition 2.8 (Geodesic is length-minimising)**.** *The path $(\mu_t)_{t\in[0,1]}$ satisfies $W_2(\mu_s, \mu_t) = |t - s| \, W_2(\mu_0, \mu_1)$ for all $s, t \in [0, 1]$.*

The Lions derivative (or L-derivative) formalises differentiation of functionals $F : \mathcal{P}_2(\mathbb{R}^d) \to \mathbb{R}$ along such geodesics. Rather than being a pointwise derivative, it is a linear functional on the tangent space. For a functional F that is Lions-differentiable, there exists a measurable function $D_\mu F(\mu) : \mathbb{R}^d \to \mathbb{R}^d$ (unique up to μ-a.e. equality) such that for any $\nu \in \mathcal{P}_2(\mathbb{R}^d)$ with optimal transport map $T = \nabla\varphi$ from μ to ν,

$$\frac{d}{dt}\Big|_{t=0} F(\mu_t) = \int_{\mathbb{R}^d} D_\mu F(\mu)(x) \cdot (T(x) - x)\, \mu(dx). \tag{2.6}$$

In other words, the directional derivative of F along a geodesic with initial velocity $v = T - \mathrm{id}$ is given by the $L^2(\mu)$ inner product of $D_\mu F(\mu)$ and v. For a comprehensive treatment, see Carmona and Delarue (2018, Vol. I, Chapter 5).

2.2.7 Law of Large Numbers in Wasserstein Distance

Theorem 2.9 (LLN Rate in W_2, Fournier–Guillin, 2015)**.** *Let $\mu \in \mathcal{P}_2(\mathbb{R}^d)$ and $X_1, X_2, \dots$ be i.i.d. with law μ. Define $\mu^N = \frac{1}{N}\sum_{i=1}^N \delta_{X_i}$.*

(i) Almost-sure convergence: $W_2(\mu^N, \mu) \to 0$ a.s. as $N \to \infty$.

(ii) Rate: If $\int_{\mathbb{R}^d} |x|^q\, \mu(dx) < \infty$ for some $q > 4$, then for $d = 1$:

$$\mathbb{E}[W_2(\mu^N, \mu)] \le C_\mu \cdot N^{-1/2}.$$

For $d \ge 3$: $\mathbb{E}[W_2(\mu^N, \mu)] \le C_{\mu,d} \cdot N^{-1/d}$, under appropriate moment conditions. (See Fournier and Guillin, 2015, Theorem 1 and Corollary 2.)

Proof of (i) in $d = 1$. By the explicit formula (5.5):

$$W_2(\mu^N, \mu)^2 = \int_0^1 |F_N^{-1}(u) - F^{-1}(u)|^2\, du.$$

For fixed u, $F_N^{-1}(u) \to F^{-1}(u)$ a.s. by the Glivenko–Cantelli theorem. Since $|F_N^{-1}(u) - F^{-1}(u)|^2 \le 4\max(|F^{-1}(u)|, |F_N^{-1}(u)|)^2$ and the right-hand side is uniformly integrable by the second moment assumption, the dominated convergence theorem gives $W_2(\mu^N, \mu)^2 \to 0$ a.s. □

Remark 2.10 (Curse of dimensionality and mean-field mitigation). The rate $N^{-1/d}$ for $d \ge 3$ highlights the curse of dimensionality in Wasserstein space: for a $d = 50$ dimensional state space, the empirical measure converges exceedingly slowly. **In practice, the mean-field assumption — coupled with the specific convex structure of the rewards and the contraction property established in Theorem 3.9 — allows particle approximation (Chapter 5) to circumvent this theoretical bottleneck.** The Wasserstein distance is used primarily as an analytical tool for stability, not as the primary object of high-dimensional statistical estimation.

Remark 2.11 (Significance for the thesis). Theorem 5.3 justifies two central approximations: (a) replacing the N-player empirical measure μ_t^N by the mean-field distribution μ_t with error $O(N^{-1/2})$ (for $d = 1$ under sufficient moments); (b) validating the block-bootstrap consistency in Theorem 9.1, where $B = 10{,}000$ samples give $O(10^{-2.5})$ estimation error.

2.2.8 Wasserstein Barycentre

Definition 2.12 (Fréchet Mean / Barycentre). Given measures $\mu_1, \ldots, \mu_K \in \mathcal{P}_2(\mathbb{R}^d)$ with weights $\lambda_k > 0$, $\sum_k \lambda_k = 1$, the Wasserstein barycentre is

$$\bar{\mu} = \arg\min_{\nu \in \mathcal{P}_2} \sum_{k=1}^{K} \lambda_k W_2(\nu, \mu_k)^2.$$

The barycentre exists and is unique when at least one μ_k is absolutely continuous (Agueh–Carlier, 2011). It is computed by the free-support Sinkhorn barycentre algorithm (Cuturi–Doucet, 2014). *Note: this concept is included for completeness but is not directly used in the subsequent chapters.*

2.3 Brownian Motion and Itô Calculus

2.3.1 Brownian Motion: Construction and Properties

Definition 2.13 (Standard Brownian Motion). A standard Brownian motion (Wiener process) $B = (B_t)_{t \geq 0}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a continuous stochastic process satisfying:

(i) $B_0 = 0$ almost surely;

(ii) Independent increments: for $0 \leq s < t$, $B_t - B_s$ is independent of $\mathcal{F}_s$;

(iii) Gaussian increments: $B_t - B_s \sim N(0, t-s)$ for all $0 \leq s < t$;

(iv) Continuous paths: $t \mapsto B_t(\omega)$ is continuous for $\mathbb{P}$-almost every ω.

Proposition 2.14 (Key path properties). *Brownian motion paths are almost surely:*

(i) Hölder continuous of order $\alpha < \frac{1}{2}$ *(but not* $\frac{1}{2}$*);*

(ii) Nowhere differentiable: $\lim_{h \to 0} \frac{B_{t+h} - B_t}{h}$ *does not exist for any* t*;*

(iii) Of unbounded variation: $\sup_\Pi \sum_k |B_{t_k} - B_{t_{k-1}}| = \infty$ *over all partitions* Π*;*

(iv) Of finite quadratic variation: $[B, B]_t = t$ *(see below).*

2.3.2 Quadratic Variation

Definition 2.15 (Quadratic Variation). The quadratic variation of a semimartingale M is the process $[M, M]_t$ defined as the limit in probability:

$$[M, M]_t = \lim_{|\Pi| \to 0} \sum_k (M_{t_k} - M_{t_{k-1}})^2,$$

where the limit is over partitions $\Pi = \{0 = t_0 < t_1 < \cdots < t_n = t\}$ as the mesh $|\Pi| = \max_k (t_k - t_{k-1}) \to 0$.

Proposition 2.16 (Quadratic variation of Brownian motion). *For standard Brownian motion* B*:* $[B, B]_t = t$ *a.s.*

Proof sketch. Let $V_n = \sum_{k=1}^{n}(B_{kh} - B_{(k-1)h})^2$ with $h = t/n$. Then $\mathbb{E}[V_n] = n \cdot h = t$ and $\mathrm{Var}(V_n) = n \cdot 2h^2 = 2t^2/n \to 0$. So $V_n \to t$ in L^2, and by Borel–Cantelli, a.s. along a subsequence. □

Remark 2.17 (Why quadratic variation matters: the Itô correction). The fact that $[B, B]_t = t \neq 0$ is the source of all the "extra terms" in stochastic calculus. For a smooth function f and a smooth path x, $f(x_t) = f(x_0) + \int f'(x_s)\,dx_s$ (ordinary chain rule). But for a Brownian path, the Taylor remainder $\frac{1}{2}f''(B_s)(dB_s)^2$ contributes $\frac{1}{2}f''(B_s)\,ds$ (not zero), because $(dB)^2 = dt$. This is Itô's formula.

2.3.3 The Itô Integral

The classical Riemann–Stieltjes integral $\int_0^T H_t\,dB_t$ is not well-defined pathwise because B has unbounded variation. Itô's construction extends it to square-integrable adapted integrands.

Definition 2.18 (Itô Integral)**.** Let $(\mathcal{F}_t)$ be the natural filtration of B, and $H = (H_t)_{t\in[0,T]}$ be $(\mathcal{F}_t)$-adapted with $\mathbb{E}[\int_0^T H_t^2\,dt] < \infty$. The Itô integral $\int_0^T H_t\,dB_t$ is the $L^2(\mathbb{P})$-limit of Riemann sums with left endpoints:

$$\int_0^T H_t\,dB_t := L^2\text{-}\lim_{n\to\infty} \sum_{k=0}^{n-1} H_{t_k}(B_{t_{k+1}} - B_{t_k}).$$

Theorem 2.19 (Itô Isometry)**.** *For adapted square-integrable H:*

$$\mathbb{E}\left[\left(\int_0^T H_t\,dB_t\right)^2\right] = \mathbb{E}\left[\int_0^T H_t^2\,dt\right]. \tag{2.7}$$

Proof sketch. For simple processes, the expectation of the square expands to $\sum_k \mathbb{E}[H_{t_k}^2]\Delta t_k$ by independence. The general case follows by approximation. □

Remark 2.20 (Significance for Theorem 3.9). The Itô isometry is used in the proof of Theorem 3.9: after deriving the Wasserstein distance ODE $dW_2^2 \leq CW_2^2\,dt + dM_t$, we take expectations and use $\mathbb{E}[M_t] = 0$ (since $M_t = \int_0^t Z_s\,dW_s^0$ is a martingale by Itô isometry, as $\mathbb{E}[\int Z_s^2\,ds] < \infty$). This eliminates the stochastic term and allows the Gronwall inequality to close.

2.3.4 Itô's Formula

Theorem 2.21 (Itô's Formula, one-dimensional)**.** *Let $f \in C^2(\mathbb{R})$ and $X_t = X_0 + \int_0^t b_s\,ds + \int_0^t \sigma_s\,dB_s$ be an Itô process. Then:*

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\,dX_s + \frac{1}{2}\int_0^t f''(X_s)\,d[X, X]_s.$$

Expanding $d[X, X]_s = \sigma_s^2\,ds$:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)b_s\,ds + \int_0^t f'(X_s)\sigma_s\,dB_s + \frac{1}{2}\int_0^t f''(X_s)\sigma_s^2\,ds. \tag{2.8}$$

Theorem 2.22 (Itô's Formula, multidimensional). *Let $f \in C^2(\mathbb{R}^d)$ and let $X_t \in \mathbb{R}^d$ satisfy $dX_t = b_t\,dt + \sigma_t\,dW_t + \sigma_0\,dW_t^0$ (where $W_t \in \mathbb{R}^d$, $W_t^0 \in \mathbb{R}^{d_0}$, and $\sigma_t \in \mathbb{R}^{d\times d}$, $\sigma_0 \in \mathbb{R}^{d\times d_0}$). Then:*

$$f(X_t) = f(X_0) + \int_0^t \nabla f(X_s)\cdot b_s\,ds + \int_0^t \nabla f(X_s)\cdot \sigma_s\,dW_s + \int_0^t \nabla f(X_s)\cdot \sigma_0\,dW_s^0 \\ + \frac{1}{2}\int_0^t \operatorname{tr}(\sigma_s\sigma_s^\top D^2 f(X_s))\,ds + \frac{1}{2}\int_0^t \operatorname{tr}(\sigma_0\sigma_0^\top D^2 f(X_s))\,ds.$$

Example 2.23 (Applying Itô to a test function). Let X_t^i be a single agent satisfying $dX_t^i = b_t\,dt + \sigma\,dW_t^i + \sigma_0\,dW_t^0$. Apply Theorem 2.22 with $f = \phi$ (a smooth test function):

$$d\phi(X_t^i) = \nabla\phi(X_t^i)\cdot b_t\,dt + \sigma\nabla\phi(X_t^i)\cdot dW_t^i + \sigma_0\nabla\phi(X_t^i)\cdot dW_t^0 \\ + \frac{\sigma^2}{2}\Delta\phi(X_t^i)\,dt + \frac{1}{2}\operatorname{tr}(\sigma_0\sigma_0^\top D^2\phi(X_t^i))\,dt.$$

Averaging over $i = 1,\dots,N$ and writing $\langle\mu_t^N,\phi\rangle = \frac{1}{N}\sum_i \phi(X_t^i)$:

$$d\langle\mu_t^N,\phi\rangle = \left\langle \mu_t^N, b_t\cdot\nabla\phi + \frac{\sigma^2}{2}\Delta\phi + \frac{1}{2}\operatorname{tr}(\sigma_0\sigma_0^\top D^2\phi)\right\rangle dt \\ + \underbrace{\frac{\sigma}{N}\sum_i \nabla\phi(X_t^i)\,dW_t^i}_{\to 0 \text{ as } N\to\infty} + \sigma_0\langle\mu_t^N,\nabla\phi\rangle\,dW_t^0.$$

The idiosyncratic term vanishes in L^2 as $N\to\infty$ (mutual independence of W^i). The common noise term survives because W_t^0 is the same for all agents. Taking $N\to\infty$ yields the Fokker–Planck SPDE of Section 2.4.

2.3.5 Itô's Product Rule and Integration by Parts

Proposition 2.24 (Itô Product Rule). *For Itô processes X_t and Y_t:*

$$d(X_tY_t) = X_t\,dY_t + Y_t\,dX_t + d[X,Y]_t,$$

where $[X,Y]_t = \int_0^t \sigma_s^X\sigma_s^Y\,ds$ is the quadratic covariation (vanishes if driven by independent Brownians).

Example 2.25 (Integration by parts for stochastic integrals). $B_t^2 = 2\int_0^t B_s\,dB_s + t$. (Verify: apply (2.8) with $f(x) = x^2$, $f'(x) = 2x$, $f''(x) = 2$.)

2.3.6 Stochastic Differential Equations

Theorem 2.26 (Existence and Uniqueness for SDEs). *Let $b : [0,T]\times\mathbb{R}^d \to \mathbb{R}^d$ and $\sigma : [0,T]\times\mathbb{R}^d \to \mathbb{R}^{d\times m}$ satisfy for some $L > 0$:*

(i) Lipschitz: $|b(t,x) - b(t,y)| + \|\sigma(t,x) - \sigma(t,y)\| \le L|x-y|$;

(ii) Linear growth: $|b(t,x)| + \|\sigma(t,x)\| \le L(1+|x|)$.

Then for any $\mathcal{F}_0$-measurable X_0 with $\mathbb{E}[|X_0|^2] < \infty$, the SDE $dX_t = b(t,X_t)\,dt + \sigma(t,X_t)\,dW_t$ has a unique strong solution $(X_t)_{t\in[0,T]}$ satisfying

$$\mathbb{E}\Big[\sup_{t\in[0,T]}|X_t|^2\Big] \le C(1+\mathbb{E}[|X_0|^2])e^{CT}.$$

Proof sketch — Picard iteration. Define $X_t^0 \equiv X_0$ and $X_t^{n+1} = X_0 + \int_0^t b(s, X_s^n)\, ds + \int_0^t \sigma(s, X_s^n)\, dW_s$. By Itô isometry and the Lipschitz condition:

$$\mathbb{E}\Big[\sup_{s\le t} |X_s^{n+1} - X_s^n|^2\Big] \le C \int_0^t \mathbb{E}\Big[\sup_{r\le s} |X_r^n - X_r^{n-1}|^2\Big]\, ds.$$

Gronwall's inequality gives $\mathbb{E}[\sup_{s\le T} |X_s^n - X_s^{n-1}|^2] \le C^n T^n / n! \to 0$. The Picard iterates converge in L^2 to a unique limit satisfying the SDE. □

2.4 Fokker–Planck SPDE

2.4.1 The Forward Equation

The Fokker–Planck equation (also called the Kolmogorov forward equation) governs the evolution of the law of a diffusion process. In the MFG setting with common noise, it becomes a stochastic PDE.

Definition 2.27 (Fokker–Planck Operator)**.** For drift $b : \mathbb{R}^d \to \mathbb{R}^d$ and diffusion $\sigma > 0$, the Fokker–Planck operator is

$$\mathcal{L}^*\phi(x) := b(x) \cdot \nabla\phi(x) + \frac{\sigma^2}{2}\Delta\phi(x), \qquad \phi \in C_c^\infty(\mathbb{R}^d).$$

This is the formal L^2-adjoint of the generator $\mathcal{L}f(x) = b(x)\cdot\nabla f(x) + \frac{\sigma^2}{2}\Delta f(x)$: $\int (\mathcal{L}f)\, d\mu = \int f(\mathcal{L}^*\mu)$ for smooth μ.

Theorem 2.28 (Fokker–Planck SPDE)**.** *Let X_t satisfy the McKean–Vlasov SDE: $dX_t = b(t, X_t, \mu_t)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0$, $\mu_t = \mathcal{L}(X_t \mid \mathcal{F}_t^{W^0})$. Then μ_t satisfies the following SPDE in the weak sense: for all $\phi \in C_c^\infty(\mathbb{R}^d)$:*

$$d\langle \mu_t, \phi\rangle = \langle \mu_t, \mathcal{L}_t^*\phi\rangle\, dt + \sigma_0 \langle \mu_t, \nabla\phi\rangle\, dW_t^0, \qquad \langle \mu_0, \phi\rangle = \int \phi(x)\, \mu_0(dx). \tag{2.9}$$

Proof. This follows directly from Example 2.23. By Itô's formula applied to $\phi(X_t)$:

$$\phi(X_t) - \phi(X_0) = \int_0^t \mathcal{L}_s^*\phi(X_s)\, ds + \sigma \int_0^t \nabla\phi(X_s)\, dW_s^i + \sigma_0 \int_0^t \nabla\phi(X_s)\, dW_s^0.$$

Take the conditional expectation given $\mathcal{F}_t^{W^0}$: the idiosyncratic term $\sigma \int \nabla\phi(X_s)\, dW_s^i$ has conditional expectation zero (since W^i is independent of W^0). The common noise term $\sigma_0 \int \nabla\phi(X_s)\, dW_s^0$ survives because W^0 is measurable with respect to $\mathcal{F}^{W^0}$. This gives the weak form (2.9). □

2.4.2 Strong Form: The SPDE in Density Form

When μ_t has a density $\rho_t(x)$ (i.e., $\mu_t(dx) = \rho_t(x)\, dx$), the SPDE (2.9) takes the strong form:

$$d\rho_t = \big[-\nabla_x \cdot (b(t, x, \mu_t)\rho_t) + \frac{\sigma^2}{2}\Delta_x \rho_t \big]\, dt + \sigma_0 \nabla_x \rho_t\, dW_t^0. \tag{2.10}$$

Remark 2.29 (Normalisation and the Kushner–Zakai distinction). Equation (2.10) is linear in ρ_t. This linearity is a consequence of two facts: (i) the SPDE describes the evolution of the conditional law of the state given the common noise, and (ii) the total mass is preserved because the drift and diffusion operators are in divergence form. Consequently, the normalisation constant remains identically one. This distinguishes the Fokker–Planck SPDE from the Kushner–Zakai equation of nonlinear filtering, which involves a stochastic normalisation factor arising from observations. The homogeneous common-noise exposure (constant σ_0) is sufficient but not necessary for this linearity; the key property is that the coefficients do not depend on the measure in a way that would introduce nonlinear feedback into the normalisation. See Chapter 4 of Carmona and Delarue (2018) for a detailed discussion.

2.4.3 Sobolev Regularity

Definition 2.30 (Sobolev Spaces). For $k \in \mathbb{N}$ and $p \in [1, \infty)$:

$$W^{k,p}(\mathbb{R}^d) = \{f \in L^p(\mathbb{R}^d) \mid \partial^\alpha f \in L^p(\mathbb{R}^d) \text{ for all } |\alpha| \leq k\},$$

with norm $\|f\|_{W^{k,p}}^p = \sum_{|\alpha| \leq k} \|\partial^\alpha f\|_{L^p}^p$. The Hilbert case $p = 2$ gives $H^k(\mathbb{R}^d) = W^{k,2}(\mathbb{R}^d)$ with inner product $\langle f, g\rangle_{H^k} = \sum_{|\alpha| \leq k} \langle \partial^\alpha f, \partial^\alpha g\rangle_{L^2}$. The dual space is $H^{-1}(\mathbb{R}^d) = (H^1(\mathbb{R}^d))^*$.

Theorem 2.31 (Fokker–Planck Well-Posedness in Sobolev Space). *Under Assumptions A1–A3 (Lipschitz drift, bounded coefficients, regularity), the SPDE (2.9) has a unique solution $\rho \in L^2(\Omega \times [0, T]; H^1(\mathbb{R}^d))$, and*

$$\mathbb{E}\Big[\sup_{t \in [0,T]} \|\rho_t\|_{L^2}^2\Big] + \mathbb{E}\left[\int_0^T \|\nabla \rho_t\|_{L^2}^2 \, dt\right] \leq C(1 + \|\rho_0\|_{H^1}^2). \tag{2.11}$$

Proof sketch. Apply the energy method: multiply (2.10) by ρ_t and integrate over $\mathbb{R}^d$. Using Itô's formula for $\|\rho_t\|_{L^2}^2$:

$$d\|\rho_t\|_{L^2}^2 = 2\langle \rho_t, d\rho_t\rangle + d[\langle \rho, \cdot\rangle, \langle \rho, \cdot\rangle]_t.$$

The deterministic part contributes $-\sigma^2 \|\nabla \rho_t\|_{L^2}^2 \, dt$ (by integration by parts) plus a term bounded by $\|b\|_\infty \|\rho_t\|_{L^2} \|\nabla \rho_t\|_{L^2} \, dt$. The stochastic part contributes $\|\sigma_0\|_F^2 \|\nabla \rho_t\|_{L^2}^2 \, dt$ (by Itô isometry). Taking expectations and applying Gronwall gives (2.11). □

2.4.4 The Gelfand Triple

Definition 2.32 (Gelfand Triple). A Gelfand triple (or rigged Hilbert space) is a triple of spaces $V \subset H \subset V^*$, where:

- V is a reflexive Banach space (e.g., $H^1(\mathbb{R}^d)$);
- H is a Hilbert space (e.g., $L^2(\mathbb{R}^d)$);
- V^* is the dual of V (e.g., $H^{-1}(\mathbb{R}^d)$);
- the inclusions are continuous and dense.

For the Fokker–Planck SPDE, the relevant Gelfand triple is $H^1(\mathbb{R}^d) \subset L^2(\mathbb{R}^d) \subset H^{-1}(\mathbb{R}^d)$. The Fokker–Planck operator $\mathcal{L}^* : H^1(\mathbb{R}^d) \to H^{-1}(\mathbb{R}^d)$ is bounded with norm $\|\mathcal{L}^*\| \leq L_b + \sigma^2$ (Lemma 4.2 of the thesis, proved via Hahn–Banach in Section 2.6). This is what makes the SPDE well-posed in H^{-1}.

2.5 Martingale Theory

2.5.1 Martingales and Stopping Times

Definition 2.33 (Martingale)**.** An adapted integrable process $(M_t)_{t\geq 0}$ is a martingale with respect to $(\mathcal{F}_t)$ if $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ $\mathbb{P}$-a.s. for all $0 \leq s \leq t$. It is a local martingale if there exist stopping times $\tau_n \uparrow \infty$ such that $(M_{t\wedge\tau_n})$ is a martingale for each n.

Theorem 2.34 (Itô integrals are local martingales)**.** *If H is adapted with $\int_0^t H_s^2\, ds < \infty$ a.s., then $M_t := \int_0^t H_s\, dB_s$ is a local martingale with $M_0 = 0$. If additionally $\mathbb{E}[\int_0^T H_s^2\, ds] < \infty$, then M_t is a square-integrable martingale.*

Theorem 2.35 (Doob's Maximal Inequality)**.** *For a non-negative submartingale (M_t):*

$$\mathbb{E}\Big[\sup_{0\leq t\leq T} M_t^2\Big] \leq 4\,\mathbb{E}[M_T^2].$$

Remark 2.36. This is used in Theorem 3.9: to bound the expected supremum

$$\mathbb{E}\big[\sup_t W_2^2(\mu_t^1, \mu_t^2)\big] \leq 4\,\mathbb{E}\big[W_2^2(\mu_T^1, \mu_T^2)\big],$$

allowing the Gronwall bound to be applied.

Theorem 2.37 (Burkholder–Davis–Gundy (BDG) Inequality)**.** *For any continuous local martingale M with $M_0 = 0$ and $p \geq 1$:*

$$c_p\,\mathbb{E}\big[[M,M]_T^{p/2}\big] \leq \mathbb{E}\Big[\sup_{t\leq T} |M_t|^p\Big] \leq C_p\,\mathbb{E}\big[[M,M]_T^{p/2}\big].$$

In particular, for $p = 2$: $\mathbb{E}[\sup_t M_t^2] \leq 4\mathbb{E}[[M,M]_T]$.

2.5.2 Martingale Representation Theorem

Theorem 2.38 (Martingale Representation, Brownian Filtration)**.** *Let $(\mathcal{F}_t)$ be the natural filtration of a d-dimensional Brownian motion W. Then every $(\mathcal{F}_t)$-martingale M has the representation*

$$M_t = M_0 + \int_0^t H_s\, dW_s$$

for a unique $(\mathcal{F}_t)$-adapted process H with $\mathbb{E}[\int_0^T |H_s|^2\, ds] < \infty$.

Remark 2.39 (Connection to BSDE theory)*.* The MRT is the existence theorem for solutions to backward SDEs. Given a terminal condition $\xi = p_T$, the MRT guarantees the existence of (q_t, r_t) such that

$$p_T = p_t - \int_t^T \partial_x H_s\, ds + \int_t^T q_s\, dW_s^i + \int_t^T r_s\, dW_s^0.$$

This is the adjoint BSDE from Chapter 1.

2.5.3 The Novikov Condition and Girsanov's Theorem

Theorem 2.40 (Girsanov's Theorem)**.** *Let $\theta : [0,T] \times \Omega \to \mathbb{R}^d$ be adapted with the Novikov condition*

$$\mathbb{E}\Big[\exp\big(\tfrac{1}{2}\int_0^T |\theta_t|^2\,dt\big)\Big] < \infty.$$

Define

$$Z_T = \exp\left(\int_0^T \theta_t\,dW_t - \frac{1}{2}\int_0^T |\theta_t|^2\,dt\right).$$

Then Z_T is the Radon–Nikodym derivative of a probability measure $\mathbb{P}^*$ with respect to $\mathbb{P}$: $d\mathbb{P}^*/d\mathbb{P} = Z_T$. Under $\mathbb{P}^*$, the process $\widetilde{W}_t = W_t - \int_0^t \theta_s\,ds$ is a Brownian motion.

Remark 2.41 (Use in the thesis). Girsanov's theorem is a standard tool for change of measure in stochastic analysis. In the MFG setting, it may be applied to remove drift terms under certain monotonicity assumptions, but the primary analysis in Theorem 3.9 uses synchronous coupling rather than Girsanov.

2.6 Sobolev Spaces and Functional Analysis

2.6.1 Sobolev Embedding Theorem

Theorem 2.42 (Sobolev Embedding)**.** *Let $\Omega \subset \mathbb{R}^d$ be bounded with smooth boundary.*

(i) If $k > d/2$, then $H^k(\Omega) \hookrightarrow C^0(\bar{\Omega})$ (continuous embedding into continuous functions);

(ii) $H^1(\Omega) \hookrightarrow L^{2^*}(\Omega)$ where $2^* = 2d/(d-2)$ for $d \geq 3$.

In our setting $\Omega = \mathbb{R}^d$, $d = 1$: $H^1(\mathbb{R}) \hookrightarrow C^{0,1/2}(\mathbb{R})$ (functions are Hölder-$\frac{1}{2}$ continuous).

2.6.2 Hahn–Banach Extension Theorem

Theorem 2.43 (Hahn–Banach)**.** *Let E be a real normed space, $F \subset E$ a subspace, and $\varphi : F \to \mathbb{R}$ a bounded linear functional. Then there exists $\widetilde{\varphi} : E \to \mathbb{R}$ with $\widetilde{\varphi}|_F = \varphi$ and $\|\widetilde{\varphi}\|_{E^*} = \|\varphi\|_{F^*}$.*

Remark 2.44 (Application in Lemma 4.2). The Hahn–Banach theorem is applied in Lemma 4.2 of the thesis to show that the Fokker–Planck operator $\mathcal{L}^*$, initially defined on smooth test functions (a dense subspace of H^1), extends to a bounded operator $\mathcal{L}^* : H^1(\mathbb{R}^d) \to H^{-1}(\mathbb{R}^d)$ with the same operator norm. This is essential for the Sobolev-space well-posedness of the SPDE.

2.6.3 Lumer–Phillips Theorem and Contraction Semigroups

Definition 2.45 (Dissipative Operator)**.** A linear operator $A : D(A) \subset H \to H$ on a Hilbert space H is dissipative if $\operatorname{Re}\langle Au, u\rangle \leq 0$ for all $u \in D(A)$.

Theorem 2.46 (Lumer–Phillips)**.** *Let A be a closed densely defined operator on a Banach space X. Then A generates a C_0-contraction semigroup (i.e., $\|e^{tA}x\| \leq \|x\|$ for all $t \geq 0$, $x \in X$) if and only if:*

(i) A *is dissipative, and*

(ii) $(\mathrm{Id} - A)$ *has dense range.*

Remark 2.47 (Application to Fokker–Planck). The Fokker–Planck operator $\mathcal{L}^*$ on $L^2(\mu^*)$ (where μ^* is the MFG equilibrium measure) is dissipative:

$$\langle \mathcal{L}^*\rho, \rho\rangle_{L^2(\mu^*)} = -\sigma^2 \|\nabla\rho\|^2_{L^2(\mu^*)} \leq 0$$

(assuming boundary terms vanish). By Lumer–Phillips, $\mathcal{L}^*$ generates a contraction semigroup, and the spectral gap $\lambda_1 > 0$ (the smallest positive eigenvalue) gives the exponential decay rate in Theorem **??**.

2.6.4 Spectral Gap and Log-Sobolev Inequality

Definition 2.48 (Spectral Gap). The spectral gap of the Fokker–Planck operator $\mathcal{L}^*$ on $L^2(\mu^*)$ is

$$\lambda_1 := \inf \left\{ \frac{-\langle \mathcal{L}^* f, f\rangle_{L^2(\mu^*)}}{\mathrm{Var}_{\mu^*}(f)} \;\middle|\; f \in D(\mathcal{L}^*),\ \mathrm{Var}_{\mu^*}(f) > 0 \right\}.$$

Theorem 2.49 (Log-Sobolev Implies Spectral Gap). *If* μ^* *satisfies the log-Sobolev inequality with constant* C_{Lip}*:*

$$H(\mu\|\mu^*) \leq C_{\mathrm{Lip}}\, I(\mu\|\mu^*) \qquad \forall \mu \in \mathcal{P}_2(\mathbb{R}^d),$$

where $H(\mu\|\mu^*) = \int \log \frac{d\mu}{d\mu^*}\, d\mu$ *and* $I(\mu\|\mu^*) = \int |\nabla \log \frac{d\mu}{d\mu^*}|^2\, d\mu$*, then* $\lambda_1 \geq 1/C_{\mathrm{Lip}}$.

Proof. The log-Sobolev inequality implies the Poincaré inequality

$$\mathrm{Var}_{\mu^*}(f) \leq C_{\mathrm{Lip}} \int |\nabla f|^2\, d\mu^*$$

(take $\mu = f^2\mu^*$, expand H to second order). The Poincaré inequality is equivalent to $\lambda_1 \geq 1/C_{\mathrm{Lip}}$. □

Remark 2.50 (Log-Sobolev in interacting particle systems). The log-Sobolev inequality is a strong structural assumption; it typically requires the equilibrium measure μ^* to satisfy Bakry–Émery criteria (e.g., strictly log-concave density). **In interacting particle systems, the equilibrium measure may exhibit heavy tails or multimodality when the potential is non-convex, which violates the LSI and requires only the weaker Poincaré inequality.** The LSI is used in the convergence-rate analysis of stochastic approximation algorithms to establish exponential convergence as a theoretical upper bound; in practice, one expects algebraic rates governed by the spectral gap. This limitation is discussed further in the future directions (Chapter 7). Theorem **??**. Recent work on Latała–Oleszkiewicz inequalities and non-convex isoperimetry (see Chapter 10) offers potential pathways to relax this assumption.

2.7 Stochastic Maximum Principle

2.7.1 The Control Problem

We now state the stochastic optimal control problem precisely. Let $A \subset \mathbb{R}^k$ be the action space (convex compact). A control $\alpha = (\alpha_t)_{t\in[0,T]}$ is admissible if it is $(\mathcal{F}_t)$-adapted and $\mathbb{E}[\int_0^T |\alpha_t|^2\, dt] < \infty$. The state $X_t^\alpha \in \mathbb{R}^d$ satisfies:

$$dX_t^\alpha = b(t, X_t^\alpha, \mu_t, \alpha_t)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0, \qquad X_0^\alpha \sim \mu_0. \tag{2.12}$$

The objective is to maximise:

$$J(\alpha) = \mathbb{E}\left[\int_0^T f(t, X_t^\alpha, \mu_t, \alpha_t)\, dt + g(X_T^\alpha, \mu_T)\right]. \tag{2.13}$$

2.7.2 The Hamiltonian

Definition 2.51 (Control Hamiltonian)**.** The Hamiltonian associated with the control problem is

$$H(t, x, \mu, p, \alpha) := p \cdot b(t, x, \mu, \alpha) + f(t, x, \mu, \alpha) \in \mathbb{R}, \tag{2.14}$$

where $p \in \mathbb{R}^d$ is the costate variable (adjoint or shadow price). The maximised Hamiltonian is

$$H^*(t, x, \mu, p) := \sup_{\alpha\in A} H(t, x, \mu, p, \alpha).$$

Under Assumption A5 (uniform concavity of H in α), H^* is attained at a unique $\alpha^*(t, x, \mu, p)$.

Example 2.52 (Linear-quadratic Hamiltonian)**.** For $b = \kappa(\bar{x}-x)+\alpha$ and $f = -x^2 - \frac{\lambda}{2}\alpha^2$:

$$H(t, x, \mu, p, \alpha) = p(\kappa(\bar{x} - x) + \alpha) - x^2 - \frac{\lambda}{2}\alpha^2.$$

Maximising over α: $\partial_\alpha H = p - \lambda\alpha = 0$, so $\alpha^*(x, \mu, p) = p/\lambda$ and $H^*(t, x, \mu, p) = p\kappa(\bar{x} - x) - x^2 + \frac{p^2}{2\lambda}$.

2.7.3 The Adjoint Process and SMP

Definition 2.53 (Adjoint BSDE)**.** The adjoint process (p_t, q_t, r_t) associated with an admissible control α and state process X_t^α satisfies the backward SDE:

$$-dp_t = \partial_x H(t, X_t^\alpha, \mu_t, p_t, \alpha_t)\, dt - q_t\, dW_t^i - r_t\, dW_t^0, \qquad p_T = \partial_x g(X_T^\alpha, \mu_T). \tag{2.15}$$

Here $p_t \in \mathbb{R}^d$ is the costate, $q_t \in \mathbb{R}^{d\times d}$ accounts for idiosyncratic noise, and $r_t \in \mathbb{R}^{d\times d_0}$ accounts for common noise.

Theorem 2.54 (Stochastic Maximum Principle)**.** *Let α^* be an optimal control for (2.13) and (p_t, q_t, r_t) the associated adjoint process. Then:*

$$H(t, X_t^*, \mu_t, p_t, \alpha_t^*) = H^*(t, X_t^*, \mu_t, p_t) \qquad \mathbb{P}\text{-a.s., a.e. } t.$$

Under Assumption A5 (strict concavity of H in α):

$$\alpha_t^* = \arg\max_{\alpha\in A} H(t, X_t^*, \mu_t, p_t, \alpha) \qquad \text{uniquely, } \mathbb{P}\text{-a.s.}$$

Proof sketch — variational method. Let $\alpha^\epsilon = \alpha^* + \epsilon(\alpha - \alpha^*)$ be a perturbation. The optimality $J(\alpha^*) \geq J(\alpha^\epsilon)$ gives, after dividing by ϵ and taking $\epsilon \to 0$:

$$\mathbb{E}\left[\int_0^T \big(\partial_\alpha H(t, X_t^*, \mu_t, p_t, \alpha_t^*)\big) \cdot (\alpha_t - \alpha_t^*)\, dt\right] \geq 0$$

for all admissible α. Since α is arbitrary, $\partial_\alpha H = 0$ $\mathbb{P}$-a.s. a.e. t. Under strict concavity (A5), this uniquely determines α_t^*. □

2.7.4 Connection to Hamilton–Jacobi–Bellman

The SMP and HJB equation are dual formulations of the same problem.

Theorem 2.55 (SMP–HJB Connection)**.** *If the value function $u(t, x, \mu)$ is $C^{1,2}$ in (t, x) and admits a Lions derivative in μ, then:*

$$p_t = D_x u(t, X_t^*, \mu_t), \qquad q_t = \sigma D_x^2 u(t, X_t^*, \mu_t), \qquad r_t = \sigma_0^\top \mathbb{E}\big[D_\mu u(t, X_t^*, \mu_t)(\widetilde{X}_t) \mid \mathcal{F}_t^{W^0}\big]. \tag{2.16}$$

Here $D_\mu u(t, x, \mu)(\widetilde{x})$ denotes the Lions derivative of u with respect to the measure argument, evaluated at the independent copy $\widetilde{X}_t$. This derivative is defined via the "lifting" of functionals on $\mathcal{P}_2(\mathbb{R}^d)$ to functionals on the Hilbert space $L^2(\Omega; \mathbb{R}^d)$; see Carmona–Delarue (2018, Vol. I, Chapter 5) for a comprehensive treatment. The HJB equation

$$-\partial_t u = H^*(t, x, \mu, D_x u) + \bar{b}(t, \mu) \cdot D_\mu u + \frac{\sigma^2 + \sigma_0^2}{2} \Delta_x u + \sigma_0^2 \operatorname{tr}\big(D_x D_\mu u\big) + \frac{\sigma_0^2}{2} \int D_\mu^2 u(\widetilde{x}, \widetilde{x})\, \mu(d\widetilde{x}),$$

$$u(T, x, \mu) = g(x, \mu),$$

where $\bar{b}(t, \mu)$ is the equilibrium aggregate drift of μ_t itself (the population analogue of the individual drift b, evaluated at the optimal control). **This is the full master equation, including the common-noise diffusion terms and the Lions-derivative mean-field coupling term $\bar{b} \cdot D_\mu u$; the latter is frequently omitted in informal expositions but is generally non-zero whenever $\bar{b}$ depends on μ, as it does in every model considered in this thesis. Chapter 7's Appendix A derivation works through this term explicitly for the linear-quadratic case, including the correction history of an earlier omission; see the note there for how the coupling term is computed in practice.** The PDE is the characterisation of the value function, and its unique solution (under Assumptions A1–A5) gives the adjoint variables via (2.16).

Remark 2.56 (Significance for the master equation)*.* Equation (2.16) reveals why the master equation $U(t, \mu)$ contains derivatives in measure: the adjoint variable r_t involves the Lions derivative $D_\mu U$, which measures how the value function changes when the measure μ_t is perturbed. **This is the analytical foundation for Chapter 6's hierarchical RL, where the slow actor learns $D_\mu U$ via the Wonham filter.**

2.8 Summary: Toolkit Reference

The following table maps each tool developed in this chapter to its first use in the thesis.

Table 2.1: Toolkit Reference

Tool	**Content**	**First used**
$\mathcal{P}_2(\mathbb{R}^d)$	Probability measures with finite 2nd moment	Ch. 3 (state space for μ^*)
Wasserstein distance	$W_2(\mu,\nu) = (\inf_\pi \int \lvert x-y\rvert^2 \pi)^{1/2}$	Ch. 3 (Theorem 3.9 LHS)
Brenier's Theorem	Optimal transport map $T = \nabla\varphi$	Ch. 3 (coupling in Lemma 3.2)
Metric space structure	$(\mathcal{P}_2, W_2)$ complete separable	Ch. 3 (Banach FPT applies)
W_2-convergence characterisation	$W_2 \to 0 \iff$ weak + 2nd moment	Ch. 3 (existence proof)
Geodesic / Lions derivative	Linear functional on $L^2(\mu)$	Ch. 8 (Lions derivative)
LLN rate (Fournier–Guillin)	$\mathbb{E}[W_2(\mu^N,\mu)] \le CN^{-1/2}$ ($d = 1$, $q > 4$)	Ch. 7 ($O(1/N)$ proof)
1D Wasserstein formula	$W_2^2 = \int (F_\mu^{-1} - F_\nu^{-1})^2$	Ch. 4 (numerical computation)
Brownian motion	Gaussian independent increments	Ch. 3 (SDE driving noise)
Quadratic variation	$[B,B]_t = t$ a.s.	Itô's formula
Itô isometry	$\mathbb{E}[(\int H dB)^2] = \mathbb{E}[\int H^2 dt]$	Ch. 3 (Step 4: $\mathbb{E}[M] = 0$)
Itô's formula	$df(X) = f' dX + \frac{1}{2} f'' d[X,X]$	Fokker–Planck derivation
SDE existence/uniqueness	Lipschitz $\Rightarrow$ unique strong solution	Ch. 3 (agent SDE)
Itô product rule	$d(XY) = X\,dY + Y\,dX + d[X,Y]$	Ch. 3 (Lemma 3.2 derivation)
Fokker–Planck SPDE	$d\langle \mu_t, \phi\rangle = \langle \mu_t, \mathcal{L}^*\phi\rangle dt + \sigma_0 \langle \mu_t, \nabla\phi\rangle dW^0$	Ch. 3 (forward equation)
Sobolev well-posedness	$\rho \in L^2(\Omega; H^1)$, energy estimate	Ch. 4 (error analysis)
Gelfand triple	$H^1 \subset L^2 \subset H^{-1}$	Ch. 4
Doob's inequality	$\mathbb{E}[\sup M^2] \le 4\mathbb{E}[M_T^2]$	Ch. 3 (sup bound)
BDG inequality	$\mathbb{E}[\sup \lvert M\rvert^p] \sim \mathbb{E}[[M,M]^{p/2}]$	Ch. 3 (moment bounds)
Martingale representation	Every martingale = Itô integral	Ch. 3 (BSDE existence)
Girsanov's theorem	Change of measure, remove drift	Ch. 3 (alternative approach)
Hahn–Banach	Norm-preserving extension	Ch. 4 (Lemma 4.2)

Chapter 3

Quantitative Well-Posedness of the Common-Noise MFG

3.1 Overview

This chapter proves Theorem 3.9 (originally labelled Theorem B.9), the central structural result of the thesis. We establish that the forward-backward stochastic differential equation (FBSDE) system (**??**) characterising the mean-field game with common noise admits a unique solution under the standing assumptions of Section 1.9. Moreover, we provide an explicit quantitative stability estimate: the solution map is a contraction in a weighted Wasserstein-2 metric on the space of measure flows, with a contraction constant that depends explicitly on the model parameters — the Lipschitz constant of the drift L_b, the Frobenius norm of the common-noise coefficient $\|\sigma_0\|_F$, and bounds on Lions derivatives. The ε-Nash property for the finite-player game is also verified.

While the existence and uniqueness of solutions to common-noise MFGs have been established under various conditions — notably by Ahuja, Ren, and Yang (2019) under convexity and weak-monotonicity, and in the comprehensive probabilistic treatment of Carmona and Delarue (2018) — the present contribution is the derivation of explicit parameter-dependent stability constants suitable for numerical analysis and reinforcement learning algorithms. These constants quantify precisely how the model parameters affect the contraction rate. **We emphasize that the contraction is established in expectation, not pathwise, and relies on the synchronous coupling of two candidate solutions sharing the identical common noise.**

The proof proceeds by formulating the MFG equilibrium as a fixed-point problem on the space of measure flows, $\Phi(\mu) = \mu$, where Φ maps a candidate measure flow to the conditional law of the optimally controlled state. Using the Stochastic Maximum Principle (Section 2.7) and the theory of McKean–Vlasov FBSDEs, we show that for any given measure flow μ, there exists a unique optimal control and corresponding state process. The key technical step is the derivation of a Wasserstein stability estimate for the map Φ in expectation, which relies on synchronous coupling of two candidate solutions, the Lions derivative calculus, and Itô's formula applied to the squared Wasserstein distance between the conditional laws. Under sufficient regularity (Assumptions 1.13–1.19), this estimate yields a contraction property in a weighted metric, from which existence and uniqueness follow by the Banach fixed-point theorem. Finally, we prove the ε-Nash property using propagation of chaos arguments.

Chapter organisation. Section 3.2 recalls the standing assumptions and notation. Section 3.3 states the main theorem. Section 3.4 defines the fixed-point map Φ. Section 3.5 contains the core stability estimate, including the synchronous coupling, Itô expansion, Lipschitz estimates, and the Gronwall argument. Section 3.6 proves the contraction in weighted Wasserstein metric. Section 3.7 concludes existence and uniqueness. Section 3.8 establishes the ε-Nash property. Section 3.9 discusses the contraction constant and its limitations. Section 3.10 provides excerpts from the Lean 4 formal verification.

3.2 Standing Assumptions and Notation

We recall the standing assumptions from Section 1.9, which are in force throughout this chapter.

Assumption 3.1 (Lipschitz drift)**.** There exists $L_b > 0$ such that for all $t \in [0, T]$, $x, x' \in \mathbb{R}^d$, $\mu, \mu' \in \mathcal{P}_2(\mathbb{R}^d)$, and $\alpha, \alpha' \in A$,

$$\|b(t, x, \mu, \alpha) - b(t, x', \mu', \alpha')\| \leq L_b\big(\|x - x'\| + W_2(\mu, \mu') + \|\alpha - \alpha'\|\big).$$

Assumption 3.2 (Lipschitz rewards)**.** The running reward f and terminal reward g are Lipschitz continuous in all arguments with constants $L_f, L_g > 0$.

Assumption 3.3 (Additional regularity)**.** The functions b, f, g are continuously differentiable in x and Lions-differentiable in μ (in the sense of Carmona and Delarue, 2018, Chapter 5). Their Lions derivatives $D_\mu b, D_\mu f, D_\mu g$ are bounded and Lipschitz continuous in x and μ. Moreover, the second-order Lions derivatives are bounded when required.

Assumption 3.4 (Bounded common noise)**.** The common-noise coefficient $\sigma_0 \in \mathbb{R}^{d \times d_0}$ satisfies $\|\sigma_0\|_F \leq M_\sigma < \infty$. The idiosyncratic coefficient $\sigma > 0$ is constant.

Assumption 3.5 (Uniform concavity of Hamiltonian)**.** The map $\alpha \mapsto H(t, x, \mu, p, \alpha)$ is uniformly concave: there exists $\lambda_H > 0$ such that for all $\alpha, \alpha' \in A$,

$$H(t, x, \mu, p, \alpha') \leq H(t, x, \mu, p, \alpha) + \partial_\alpha H(t, x, \mu, p, \alpha) \cdot (\alpha' - \alpha) - \frac{\lambda_H}{2}\|\alpha' - \alpha\|^2.$$

Assumption 3.6 (Regular terminal reward)**.** The terminal reward g is C^1 in x and Lions-differentiable in μ, with bounded derivatives.

Assumption 3.7 (Initial condition)**.** The initial distribution $\mu_{\text{init}} \in \mathcal{P}_2(\mathbb{R}^d)$ has finite fourth moment: $\int_{\mathbb{R}^d} |x|^4 \, \mu_{\text{init}}(dx) < \infty$.

Notation. We work on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in [0,T]}, \mathbb{P})$ described in Section 1.2.1. The space of measure flows is denoted by $\mathcal{M} := D([0, T]; \mathcal{P}_2(\mathbb{R}^d))$, the Skorokhod space of càdlàg measure-valued processes adapted to the common-noise filtration $(\mathcal{F}_t^{W^0})$. For $\mu \in \mathcal{M}$ and $\lambda > 0$, the weighted Wasserstein-2 norm is

$$\|\mu\|_{\lambda,T} := \sup_{t \in [0,T]} e^{-\lambda t}\mathbb{E}\big[W_2(\mu_t, \delta_0)^2\big]^{1/2},$$

where δ_0 is the Dirac measure at zero. For two measure flows $\mu^1, \mu^2 \in \mathcal{M}$, the distance is

$$\|\mu^1 - \mu^2\|_{\lambda,T} := \sup_{t \leq T} e^{-\lambda t}\mathbb{E}\big[W_2(\mu_t^1, \mu_t^2)^2\big]^{1/2}.$$

Remark 3.8 (Weighted norm). This weighted norm is equivalent to the standard norm used in Carmona and Delarue (2018) but is more convenient for extracting contraction factors. The exponential discounting allows us to absorb the Gronwall growth factor.

3.3 Statement of the Main Theorem

Theorem 3.9 (Quantitative Well-Posedness). *Under Assumptions 1.13–1.19, there exists a unique solution $(\mu^*, \alpha^*, X^*, p, q, r)$ to the FBSDE system (**??**). Moreover, for any two measure flows $\mu^1, \mu^2 \in \mathcal{M}$, the map Φ defined in (3.6) below satisfies the stability estimate in expectation:*

$$\mathbb{E}\big[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2\big] \le C_0 \int_0^t e^{C(t-s)} \mathbb{E}\big[W_2(\mu^1_s, \mu^2_s)^2\big]\, ds, \tag{3.1}$$

where the constants are

$$C = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}, \qquad C_0 = L_b + C_\alpha L_b,$$

with C_{reg} depending on the Lipschitz constants of the Lions derivatives of b, f, g and the bounds from the adjoint BSDE, and C_α is the Lipschitz constant of the optimal control with respect to state, costate, and measure. Consequently, for any $\lambda > C/2$, the map Φ is a contraction in the weighted norm $\|\cdot\|_{\lambda,T}$:

$$\|\Phi(\mu^1) - \Phi(\mu^2)\|_{\lambda,T} \le \rho\|\mu^1 - \mu^2\|_{\lambda,T}, \qquad \rho = \sqrt{\frac{C_0}{2\lambda - C}} < 1. \tag{3.2}$$

The unique fixed point $\mu^* = \Phi(\mu^*)$ is the MFG equilibrium measure flow. Finally, the ε-Nash property holds: if all N players adopt the MFG strategy α^*, the strategy profile is an ε_N-Nash equilibrium of the N-player game with $\varepsilon_N = O(N^{-1/2})$.

Remark 3.10 (On the constant C_{reg}). The constant C_{reg} is not fully explicit; it encapsulates bounds on Lions derivatives and BSDE estimates. However, in specific models (e.g., the linear-quadratic MFG), all constants can be computed explicitly in terms of the model parameters. For general nonlinear MFGs, C_{reg} can be bounded by quantities involving the Lipschitz constants of the Lions derivatives and the operator norms of the associated linearised BSDEs.

Remark 3.11 (Expectation vs. pathwise). The contraction is established at the level of expected Wasserstein distance. The first-order stochastic terms cancel in expectation due to the martingale property, but they do not cancel pathwise. Consequently, the stability estimate (3.1) holds for the expectation, not almost surely.

3.4 The Fixed-Point Map Φ

For a given measure flow $\mu = (\mu_t)_{t\in[0,T]} \in \mathcal{M}$, we consider the optimal control problem of a representative agent who takes μ as given. The agent's state process X^μ satisfies the McKean–Vlasov SDE

$$dX_t^\mu = b(t, X_t^\mu, \mu_t, \alpha_t)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0, \qquad X_0^\mu \sim \mu_{\text{init}}, \tag{3.3}$$

and the objective is to maximise

$$J^\mu(\alpha) = \mathbb{E}\left[\int_0^T f(t, X_t^\mu, \mu_t, \alpha_t)\, dt + g(X_T^\mu, \mu_T)\right]. \tag{3.4}$$

By the Stochastic Maximum Principle (Theorem 2.20), an optimal control α^μ exists and is unique under Assumptions 1.13–1.17. It is characterised by the forward-backward system

$$\begin{aligned} dX_t^\mu &= b(t, X_t^\mu, \mu_t, \alpha_t^\mu)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0, \qquad X_0^\mu \sim \mu_{\text{init}}, \\ -dp_t^\mu &= \partial_x H(t, X_t^\mu, \mu_t, p_t^\mu, \alpha_t^\mu)\, dt - q_t^\mu\, dW_t^i - r_t^\mu\, dW_t^0, \qquad p_T^\mu = \partial_x g(X_T^\mu, \mu_T), \\ \alpha_t^\mu &= \arg\max_{\alpha \in A} H(t, X_t^\mu, \mu_t, p_t^\mu, \alpha), \end{aligned} \tag{3.5}$$

where $H(t, x, \mu, p, \alpha) = p \cdot b(t, x, \mu, \alpha) + f(t, x, \mu, \alpha)$.

The solution to (3.5) exists and is unique for each fixed $\mu \in \mathcal{M}$; this follows from standard results for McKean–Vlasov FBSDEs with Lipschitz coefficients (see, e.g., Carmona & Delarue, 2018, Vol. I, Chapter 4, and the monotone functional approach of Ahuja, Ren, and Yang, 2019). We denote the unique optimal state process by X^μ and define the map

$$\Phi(\mu)_t := \mathcal{L}\big(X_t^\mu \mid \mathcal{F}_t^{W^0}\big), \qquad t \in [0, T]. \tag{3.6}$$

That is, $\Phi(\mu)$ is the measure flow obtained as the conditional law of the optimally controlled state given the common noise. By construction, $\Phi(\mu) \in \mathcal{M}$. An MFG equilibrium is precisely a fixed point of Φ.

3.5 Stability Estimate for the Map Φ

The core of the proof is to show that Φ satisfies the Lipschitz-type estimate (3.1) in expectation. Let $\mu^1, \mu^2 \in \mathcal{M}$ be two candidate measure flows. We denote by $(X^1, p^1, q^1, r^1, \alpha^1)$ and $(X^2, p^2, q^2, r^2, \alpha^2)$ the corresponding solutions to (3.5) with μ replaced by μ^1 and μ^2, respectively. The goal is to bound $\mathbb{E}[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2]$.

3.5.1 Synchronous Coupling

A central idea is to couple the two systems on the same probability space by using the same realisations of the idiosyncratic noise W^i and, crucially, the same common noise W^0. This synchronous coupling ensures that differences arising from the common noise partially cancel in expectation when comparing the two state processes.

Concretely, let $(\Omega, \mathcal{F}, \mathbb{P})$ support independent Brownian motions W^i and W^0, and let the initial conditions X_0^1, X_0^2 be coupled optimally, i.e., their joint distribution achieves $W_2(\mu_{\text{init}}, \mu_{\text{init}}) = 0$ (they are equal in law). We solve the two systems (3.5) on this same probability space with the same driving noises. Then, for each t, the conditional laws $\Phi(\mu^1)_t$ and $\Phi(\mu^2)_t$ are the distributions of X_t^1 and X_t^2 conditioned on $\mathcal{F}_t^{W^0}$. The Wasserstein distance between these conditional laws can be bounded by the L^2-distance between the random variables under the optimal coupling of their conditional distributions. Since we have already coupled the processes synchronously, we can use the joint distribution of (X_t^1, X_t^2) to obtain a bound:

$$\mathbb{E}\big[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2\big] \le \mathbb{E}\big[|X_t^1 - X_t^2|^2\big]. \tag{3.7}$$

This inequality is standard: the Wasserstein distance between two laws is bounded above by the L^2-distance between any coupling of random variables with those laws.

3.5.2 Itô Expansion of the Squared Difference

We now study the dynamics of the difference process $\Delta X_t := X_t^1 - X_t^2$. From the forward SDEs,

$$d(\Delta X_t) = \big[b(t, X_t^1, \mu_t^1, \alpha_t^1) - b(t, X_t^2, \mu_t^2, \alpha_t^2)\big]\, dt,$$

since the noise terms $\sigma\, dW_t^i$ and $\sigma_0\, dW_t^0$ cancel identically under synchronous coupling. **Crucially, this cancellation holds pathwise for the difference of the two processes, not just in expectation. The stochastic terms disappear entirely from the equation for ΔX_t.** Applying Itô's formula to $|\Delta X_t|^2$, we obtain

$$d|\Delta X_t|^2 = 2\Delta X_t \cdot \big[b(t, X_t^1, \mu_t^1, \alpha_t^1) - b(t, X_t^2, \mu_t^2, \alpha_t^2)\big]\, dt. \tag{3.8}$$

Notice that there is no Itô correction term because the diffusion coefficients are constant and identical for both processes, so their difference has zero quadratic variation. This is a crucial simplification resulting from the synchronous coupling and the assumption of constant σ, σ_0.

3.5.3 Lipschitz Estimates and Gronwall Setup

Using the Lipschitz property of b (Assumption 3.1), we have

$$|b(t, X_t^1, \mu_t^1, \alpha_t^1) - b(t, X_t^2, \mu_t^2, \alpha_t^2)| \le L_b\big(|\Delta X_t| + W_2(\mu_t^1, \mu_t^2) + |\alpha_t^1 - \alpha_t^2|\big).$$

Substituting into (3.8) and using Young's inequality $2ab \le a^2 + b^2$, we get

$$d|\Delta X_t|^2 \le \big[(2L_b + L_b^2)|\Delta X_t|^2 + L_b W_2(\mu_t^1, \mu_t^2)^2 + L_b|\alpha_t^1 - \alpha_t^2|^2\big]\, dt. \tag{3.9}$$

To close the estimate, we need to bound the difference in controls $|\alpha_t^1 - \alpha_t^2|$ in terms of the state and measure differences. This is where the regularity of the Hamiltonian and the adjoint processes enters.

3.5.4 Control Difference via SMP and Lions Derivatives

From the optimality condition (3.5) and the uniform concavity of H (Assumption 1.17), the optimal control is a Lipschitz function of the state x, the costate p, and the measure μ. Specifically, by the implicit function theorem applied to $\partial_\alpha H = 0$, there exists a constant $C_\alpha > 0$ such that

$$|\alpha_t^1 - \alpha_t^2| \le C_\alpha\big(|X_t^1 - X_t^2| + |p_t^1 - p_t^2| + W_2(\mu_t^1, \mu_t^2)\big). \tag{3.10}$$

The rigorous justification of this Lipschitz dependence relies on the Lions differentiability of b and f with respect to μ (Assumption 3.3). By Carmona and Delarue (2018, Proposition 5.35), if a functional is Lions-differentiable with bounded derivative, it is Lipschitz in the Wasserstein distance. The constant C_α can be bounded explicitly in terms of λ_H, L_b, and the bounds on the Lions derivatives. A detailed derivation is provided in Appendix A.

3.5.5 Adjoint Process Difference

Next, we need an estimate for the difference of the adjoint processes $\Delta p_t := p_t^1 - p_t^2$. The BSDEs for p^1 and p^2 are

$$-dp_t^j = \partial_x H(t, X_t^j, \mu_t^j, p_t^j, \alpha_t^j)\, dt - q_t^j\, dW_t^i - r_t^j\, dW_t^0, \qquad j = 1, 2,$$

with terminal conditions $p_T^j = \partial_x g(X_T^j, \mu_T^j)$. Taking the difference, we obtain a linear BSDE for Δp_t:

$$-d(\Delta p_t) = \big[A_t \Delta X_t + B_t \Delta p_t + C_t \star (\mu_t^1 - \mu_t^2) + D_t(\alpha_t^1 - \alpha_t^2)\big]\, dt - \Delta q_t\, dW_t^i - \Delta r_t\, dW_t^0, \tag{3.11}$$

where the coefficients A_t, B_t, D_t involve derivatives of $\partial_x H$ with respect to x, p, α respectively, and are bounded thanks to Assumption 3.3. The term $C_t \star (\mu_t^1 - \mu_t^2)$ denotes the Lions derivative of $\partial_x H$ with respect to the measure, evaluated in the direction of the difference of measures. By Assumption 3.3, this Lions derivative is a bounded linear operator on the tangent space, and its operator norm is bounded by a constant L_μ. Consequently,

$$\|C_t \star (\mu_t^1 - \mu_t^2)\| \le L_\mu W_2(\mu_t^1, \mu_t^2). \tag{3.12}$$

Standard BSDE estimates (see, e.g., El Karoui et al., 1997, or Zhang, 2017, Theorem 4.3.1) yield

$$\begin{aligned} &\mathbb{E}\left[\sup_{t\in[0,T]} |\Delta p_t|^2 + \int_0^T \big(|\Delta q_t|^2 + |\Delta r_t|^2\big)\, dt\right] \\ &\le C_{\mathrm{BSDE}}\, \mathbb{E}\left[|\Delta X_T|^2 + \int_0^T \big(|\Delta X_t|^2 + |\alpha_t^1 - \alpha_t^2|^2 + W_2(\mu_t^1, \mu_t^2)^2\big)\, dt\right]. \end{aligned} \tag{3.13}$$

The constant C_{BSDE} depends on the bounds of the coefficients A_t, B_t, D_t, the Lipschitz constant L_μ, the time horizon T, and crucially, the norm of the common-noise volatility $\|\sigma_0\|_F$ (through the martingale representation terms). A precise expression is given in Appendix A.

3.5.6 Combining the Estimates

We now have a coupled system of inequalities for $\mathbb{E}[|\Delta X_t|^2]$ and $\mathbb{E}[|\Delta p_t|^2]$. Using (3.10) to bound the control difference, and plugging into (3.9) and (3.13), we can eliminate $|\alpha_t^1 - \alpha_t^2|$ and obtain a closed differential inequality for

$$y_t := \mathbb{E}\big[|\Delta X_t|^2\big] + \mathbb{E}\big[|\Delta p_t|^2\big].$$

The algebraic manipulations, detailed in Appendix A, yield

$$\frac{d}{dt} y_t \le C_1 y_t + C_2\, \mathbb{E}\big[W_2(\mu_t^1, \mu_t^2)^2\big], \tag{3.14}$$

where

$$C_1 = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\mathrm{reg}}, \qquad C_2 = L_b + C_\alpha L_b + C_{\mathrm{BSDE}} L_\mu.$$

Here C_{reg} encapsulates the regularity constants from the Lions derivatives and the BSDE bounds. The term $2\|\sigma_0\|_F^2$ arises from the dependence of the BSDE constant C_{BSDE} on the common-noise volatility; specifically, the martingale representation part of the BSDE estimate involves the quadratic variation of the common noise, contributing a term proportional to $\|\sigma_0\|_F^2$. The explicit derivation is provided in Appendix A.

3.5.7 The Wasserstein Stability Estimate

By Gronwall's inequality, (3.14) implies

$$y_t \le C_2 \int_0^t e^{C_1(t-s)} \mathbb{E}\big[W_2(\mu_s^1, \mu_s^2)^2\big]\, ds.$$

Recalling that y_t bounds $\mathbb{E}[|\Delta X_t|^2]$ from above, and using (3.7), we obtain precisely the stability estimate (3.1) with $C = C_1$ and $C_0 = C_2$:

$$\mathbb{E}\big[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2\big] \le C_0 \int_0^t e^{C(t-s)} \mathbb{E}\big[W_2(\mu_s^1, \mu_s^2)^2\big]\, ds.$$

Remark 3.12 (Pathwise vs. expected contraction). The inequality above holds for the expected squared Wasserstein distance. The first-order stochastic terms from the common noise cancel in expectation because the martingale $M_t = \int_0^t Z_s\, dW_s^0$ has zero mean. This is why the contraction is established in the weighted norm involving an expectation.

3.6 Contraction in Weighted Wasserstein Metric

We now show that Φ is a contraction in the weighted norm $\|\cdot\|_{\lambda,T}$ for a suitably chosen λ. Take the supremum over $t \in [0,T]$ of both sides of (3.1) multiplied by $e^{-2\lambda t}$:

$$\begin{aligned}
\|\Phi(\mu^1) - \Phi(\mu^2)\|_{\lambda,T}^2 &= \sup_{t\le T} e^{-2\lambda t} \mathbb{E}\big[W_2(\Phi(\mu^1)_t, \Phi(\mu^2)_t)^2\big] \\
&\le C_0 \sup_{t\le T} e^{-2\lambda t} \int_0^t e^{C(t-s)} \mathbb{E}\big[W_2(\mu_s^1, \mu_s^2)^2\big]\, ds \\
&\le C_0 \sup_{t\le T} \int_0^t e^{-2\lambda(t-s)} e^{(C-2\lambda)s} e^{-2\lambda s} \mathbb{E}\big[W_2(\mu_s^1, \mu_s^2)^2\big]\, ds \\
&\le C_0 \left(\int_0^T e^{-(2\lambda - C)u}\, du \right) \|\mu^1 - \mu^2\|_{\lambda,T}^2,
\end{aligned}$$

where we used the change of variables $u = t - s$ and the fact that $e^{-2\lambda s}\mathbb{E}[W_2^2] \le \|\mu^1 - \mu^2\|_{\lambda,T}^2$. The integral

$$I(\lambda) := C_0 \int_0^T e^{-(2\lambda - C)u}\, du = \frac{C_0}{2\lambda - C}\big(1 - e^{-(2\lambda - C)T}\big) \le \frac{C_0}{2\lambda - C}.$$

Thus, for any $\lambda > C/2$, we have $I(\lambda) < 1$ provided λ is chosen large enough. More precisely, choosing $\lambda > \frac{C}{2} + \frac{C_0}{2}$ ensures the contraction factor $\rho = \sqrt{C_0/(2\lambda - C)}$ is strictly less than 1. This makes Φ a strict contraction on the complete metric space $(\mathcal{M}, \|\cdot\|_{\lambda,T})$.

3.7 Existence and Uniqueness

Since $\mathcal{M}$ equipped with the weighted Wasserstein norm is a complete metric space (the completeness follows from the completeness of $\mathcal{P}_2(\mathbb{R}^d)$ and standard properties of Skorokhod space; see Billingsley, 1999), the Banach fixed-point theorem guarantees the existence of a unique fixed point $\mu^* \in \mathcal{M}$ such that $\Phi(\mu^*) = \mu^*$. This fixed point is the MFG equilibrium measure flow. The corresponding processes $(X^*, p^*, q^*, r^*, \alpha^*)$ obtained from solving (3.5) with $\mu = \mu^*$ constitute the unique solution to the full FBSDE system (**??**).

Remark 3.13 (Local vs. global contraction). The contraction constant ρ depends on the time horizon T through the constants C and C_0. For large T, the condition $\lambda > C/2$ may require a large λ, but the contraction still holds. In some cases, the contraction is only local in time; however, under strengthened convexity conditions (e.g., strong monotonicity of the cost), the contraction can be global.

3.8 The ε-Nash Property

Finally, we verify that the MFG strategy α^* provides an approximate Nash equilibrium for the N-player game. Consider the N-player system where all agents except agent i follow the MFG strategy α^*, while agent i uses an arbitrary admissible control α^i. Let $\mu_t^{N,-i}$ be the empirical measure of the other $N-1$ agents. By the conditional propagation of chaos (see Carmona & Delarue, 2018, Vol. II, Theorem 6.16), the empirical measure $\mu_t^{N,-i}$ converges to the MFG measure flow μ^* in the sense that

$$\mathbb{E}\big[W_2(\mu_t^{N,-i}, \mu_t^*)^2\big] \leq \frac{C_{\mathrm{PoC}}}{N},$$

where the constant C_{PoC} depends on the model parameters (Lipschitz constants, moments) but not on N. Moreover, the state of agent i under control α^i and the MFG state under α^* can be coupled such that their distance is controlled by the measure approximation error.

Using the Lipschitz properties of the rewards and the stability of the FBSDE system (essentially the estimate (3.1) applied to the finite-N empirical measure), one can show that the difference in objective values satisfies

$$\sup_{\alpha^i} \big(J_i(\alpha^i, \alpha^{-i,*}) - J_i(\alpha^*, \alpha^{-i,*})\big) \leq \varepsilon_N,$$

with $\varepsilon_N = O(N^{-1/2})$. Thus, the symmetric strategy profile $(\alpha^*, \ldots, \alpha^*)$ is an ε_N-Nash equilibrium. This completes the proof of Theorem 3.9.

Remark 3.14 (Verification of propagation of chaos conditions). The proof relies on the conditions of Carmona and Delarue (2018, Theorem 6.16), which are satisfied under Assumptions 1.13–1.19. In particular, the Lipschitz property of the drift and the boundedness of the diffusion coefficients ensure the conditional propagation of chaos holds.

3.9 Discussion of the Contraction Constant

The explicit constant $C = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\mathrm{reg}}$ appearing in the stability estimate (3.1) merits comment.

- The term $2L_b^2$ arises from the Lipschitz property of the drift.
- The term $2\|\sigma_0\|_F^2$ reflects the second-order contribution of the common noise to the Wasserstein dynamics; note that the first-order stochastic term cancels in expectation due to the martingale property, leaving only the effect of σ_0 through the BSDE estimates and the control difference.

- The remainder C_{reg} encapsulates the regularity constants from the Lions derivatives of b, f, g and the bounds from the adjoint BSDE. In the special case of the linear-quadratic MFG (Example 1.6), one can compute C_{reg} explicitly in terms of the model parameters $\kappa, \lambda, \sigma, \sigma_0$. For general nonlinear MFGs, C_{reg} is not fully explicit but can be bounded by quantities involving the Lipschitz constants of the Lions derivatives.

Table 3.1: Components of the Contraction Constant

Term	**Expression**	**Origin**
Drift Lipschitz	$2L_b^2$	From ΔX_t Gronwall step
Common noise	$2\|\sigma_0\|_F^2$	From BSDE constant dependence
Regularity	C_{reg}	Lions derivatives, BSDE bounds
Control Lipschitz	C_α	Implicit function theorem on H

Limitations. The contraction estimate relies on the stability of optimal transport maps, which requires absolute continuity of measures and bounded densities (Assumption 3.3). In the particle-method implementation of Chapter 5, the empirical measure is discrete, and this assumption is violated. **The theoretical results should be interpreted as establishing a well-posedness baseline under idealised conditions; practical implementation requires regularisation (e.g., kernel density estimation or entropic smoothing) to approximate the required regularity. The error introduced by such regularisation is not quantified here and remains an important direction for future work.**

3.10 Formal Verification in Lean 4

The key estimates of this chapter have been formalised using the Lean 4 interactive theorem prover, targeting the Gronwall estimate, the contraction property in weighted Wasserstein metric, and the stability constant derivation. **As detailed in the provenance caveat at the start of Appendix B, these proofs were written and reviewed for mathematical correctness following standard Mathlib idioms, but were not compiled against a live Lean 4/Mathlib installation while writing this thesis, since no such toolchain was available in that environment; they should not be treated as machine-verified until built successfully via `lake build` (the supplementary repository's CI workflow does this on every push).** The full Lean 4 code is available in the supplementary repository.

Listing 3.1: Lean 4 formalisation of the Gronwall estimate underlying the contraction proof.

```
-- Lean 4 formalization of the contraction estimate (Theorem 3.9)
import Mathlib.Analysis.SpecialFunctions.Exp
import Mathlib.Analysis.ODE.Gronwall
import Mathlib.MeasureTheory.Measure.ProbabilityMeasure
import Mathlib.Tactic.Linarith

```

```
open Real MeasureTheory ProbabilityMeasure

variable {C1 C2 : Real} (hC1 : 0 < C1) (hC2 : 0 <= C2)

/-- Gronwall inequality for differential inequality y' <= C1 * y + C2 -/
theorem gronwall_estimate {y : Real -> Real} {C1 C2 : Real}
    (hC1 : 0 < C1) (hC2 : 0 <= C2)
    (h_deriv : forall t >= 0, HasDerivAt y (deriv y t) t)
    (h_bound : forall t >= 0, deriv y t <= C1 * y t + C2)
    (hy0 : y 0 = 0) :
    forall t >= 0, y t <= C2 * (Real.exp (C1 * t) - 1) / C1 := by
  intro t ht
  let z (t : Real) : Real := y t - C2 / C1 * (Real.exp (C1 * t) - 1)
  have hz0 : z 0 = 0 := by simp [z, hy0, Real.exp_zero, sub_self, mul_zero]
  -- [excerpt truncated for typesetting -- continues with the derivative
  --  bound on z, then le_of_deriv_le to conclude z t <= 0 for all t >= 0,
  --  giving the stated bound on y. The full proof, together with the
  --  provenance caveat governing its verification status, is reproduced
  --  in Appendix B.]
  exact gronwall_estimate_aux hC1 hC2 h_deriv h_bound hy0 t ht
```

Note: the excerpt above is abbreviated for typesetting; the complete proof, together with the provenance caveat explaining that it has not yet been compiled against Mathlib, is provided in full in Appendix B.

Chapter 4

Numerical Approximation of the Common-Noise MFG System

4.1 Overview

This chapter develops numerical methods for approximating the solution to the forward-backward stochastic differential equation (FBSDE) system (**??**) that characterises the mean-field game equilibrium with common noise. The primary theoretical result is Theorem 4.8, which establishes the convergence rate of a combined Euler–Maruyama / particle discretisation scheme under sufficient smoothness assumptions. A CFL-type stability condition is derived for the coupled forward-backward system, ensuring that the discrete approximation remains well-behaved as the mesh is refined.

The numerical challenge is threefold. First, the forward component consists of a McKean–Vlasov SDE for the representative agent's state, coupled to the population measure flow. Second, the measure flow itself evolves according to a stochastic partial differential equation (the Fokker–Planck SPDE). Third, the adjoint process satisfies a backward SDE whose terminal condition depends on the final state and the terminal measure. A full discretisation must handle all three components simultaneously while preserving the fixed-point structure that defines the equilibrium.

We adopt a particle method for the population measure: the law of the state is approximated by an empirical measure of M simulated trajectories. The state SDE and the adjoint BSDE are discretised using the Euler–Maruyama scheme. The Fokker–Planck SPDE is discretised via an explicit finite-difference scheme on a spatial grid, or equivalently via the empirical measure of the particle system. The error analysis focuses on the weak convergence of the discretised measure flow to the true equilibrium measure, measured in the Wasserstein-2 distance.

Prior work. The numerical analysis of McKean–Vlasov FBSDEs has been studied by Chassagneux, Crisan, and Delarue (2019), who established convergence of a particle method for a class of mean-field FBSDEs without common noise. The CEMRACS 2017 project (see Angiuli et al., 2017) investigated numerical probabilistic methods for MFGs, including tree-based and regression approaches. **The present chapter extends these techniques to the common-noise setting, where the measure flow is stochastic and the Fokker–Planck equation becomes an SPDE.** The finite-difference analysis builds on the work of Achdou and Porretta (2016) for deterministic MFG PDEs and on Lord, Powell, and Shardlow (2014) for SPDEs. The weak error analysis for particle discretisations of McKean–Vlasov SDEs follows the framework of Bencheikh (2020). The BSDE regression method is based on the foundational work of Gobet, Lemor, and Warin (2005) and Bouchard and Touzi (2004).

Limitations and assumptions. The convergence results in this chapter rely on strong smoothness assumptions (coefficients of class $C^{2,4}$) that may not hold in realistic financial models with non-smooth transaction costs or jumps. The particle method suffers from the curse of dimensionality, with convergence rate $O(M^{-1/d})$ for $d \geq 3$. The analysis provides theoretical guarantees under idealised conditions; practical implementation may require regularisation and careful tuning.

Chapter organisation. Section 4.2 sets up the discrete-time approximation of the FBSDE system. Section 4.3 analyses the particle approximation of the measure flow. Section 4.4 derives the weak error estimate for the Euler–Maruyama scheme applied to the state and adjoint processes. Section 4.5 presents the finite-difference scheme for the Fokker–Planck SPDE and establishes its stability and convergence. Section 4.6 combines these estimates to prove Theorem 4.8, giving the overall convergence rate. Section 4.7 derives the CFL condition for the coupled system. Section 4.8 presents numerical experiments validating the theoretical rates on the linear-quadratic MFG benchmark. Section 4.9 concludes with a discussion of implementation considerations.

4.2 Discretisation of the FBSDE System

We recall the FBSDE system (**??**) characterising the MFG equilibrium: the unique solution $(\mu^*, \alpha^*, X^*, p, q, r)$ exists by Theorem 3.9. For numerical approximation, we fix a time grid $0 = t_0 < t_1 < \cdots < t_N = T$ with uniform step size $\Delta t = T/N$, and a spatial grid for the measure flow with mesh size Δx. We denote by X^n, p^n, μ^n the approximations at time t_n.

4.2.1 Discrete Forward SDE

The state process satisfies $dX_t = b(t, X_t, \mu_t, \alpha_t)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0$. The Euler–Maruyama discretisation is

$$X^{n+1} = X^n + b(t_n, X^n, \mu^n, \alpha^n)\, \Delta t + \sigma \Delta W_n^i + \sigma_0 \Delta W_n^0, \tag{4.1}$$

where $\Delta W_n^i \sim N(0, \Delta t\, I_d)$ and $\Delta W_n^0 \sim N(0, \Delta t\, I_{d_0})$ are independent Gaussian increments. The initial condition X^0 is sampled from μ_{init}.

4.2.2 Discrete Adjoint BSDE

The adjoint process satisfies $-dp_t = \partial_x H(t, X_t, \mu_t, p_t, \alpha_t)\, dt - q_t\, dW_t^i - r_t\, dW_t^0$, $p_T = \partial_x g(X_T, \mu_T)$. The Euler–Maruyama scheme for BSDEs proceeds backward in time. Given terminal values $p^N = \partial_x g(X^N, \mu^N)$, we compute for $n = N-1, \ldots, 0$:

$$p^n = \mathbb{E}_n\big[p^{n+1} + \partial_x H(t_n, X^n, \mu^n, p^n, \alpha^n)\, \Delta t\big], \tag{4.2}$$

where $\mathbb{E}_n[\cdot]$ denotes the conditional expectation given the information at time t_n. In practice, this conditional expectation is approximated by nonparametric regression on a set of basis functions. Following Gobet, Lemor, and Warin (2005), we project the random variable inside the expectation onto a finite-dimensional function space spanned by basis functions $\{\psi_k(X^n)\}_{k=1}^{K}$. The regression coefficients are estimated by least squares on a set of M simulated paths. The processes q^n and r^n can be obtained from the martingale representation theorem, but they are not explicitly needed for the optimal control if we use a direct regression approach for p^n.

4.2.3 Discrete Optimality Condition

At each time step, the control is determined by maximising the Hamiltonian:

$$\alpha^n = \arg\max_{\alpha \in A} H(t_n, X^n, \mu^n, p^n, \alpha). \tag{4.3}$$

For the LQ case, this yields the explicit formula $\alpha^n = p^n/\lambda$.

4.2.4 Discrete Consistency Condition

The population measure μ^n must equal the conditional law of X^n given the common noise. In the discrete setting, this is enforced by solving the fixed-point equation $\mu = \Phi^{\Delta}(\mu)$, where Φ^{Δ} is the discrete analogue of the map Φ defined in Chapter 3. The fixed-point iteration is performed using a particle system or a grid-based approximation.

4.3 Particle Approximation of the Measure Flow

To approximate the measure flow μ_t, we simulate M independent copies of the agent's state process, all driven by the same common noise W^0 but independent idiosyncratic noises $W^{i,m}$. Let $X^{m,n}$ denote the state of the m-th particle at time t_n. The empirical measure is

$$\mu^{M,n} = \frac{1}{M} \sum_{m=1}^{M} \delta_{X^{m,n}}. \tag{4.4}$$

By the law of large numbers in Wasserstein distance (Theorem 5.3), $\mu^{M,n}$ converges to the true conditional law μ^n as $M \to \infty$. The rate of convergence is $O(M^{-1/2})$ in one dimension (under finite moments of order $q > 4$) and $O(M^{-1/d})$ in dimension $d \geq 3$. In the analysis that follows, we assume that the number of particles M is sufficiently large so that the particle approximation error is dominated by the time and space discretisation errors. **For the LQ benchmark in Section 4.8, we use $M = 10^5$ particles to ensure negligible statistical error.**

The particle system interacts through the empirical measure: the drift and the Hamiltonian are evaluated at $\mu^{M,n}$. This yields a fully discrete, simulatable system. The fixed-point iteration to find the equilibrium measure μ^* is performed by iterating the particle system until the empirical measure stabilises.

Remark 4.1 (Kernel Density Estimation). For higher-dimensional problems where the particle curse of dimensionality is severe, one may replace the empirical measure with a kernel density estimator $\hat{\mu}^{M,n}(x) = \frac{1}{Mh^d} \sum_{m=1}^{M} K\left(\frac{x - X^{m,n}}{h}\right)$, where K is a smooth kernel and $h > 0$ is the bandwidth. This smooths the measure and can improve convergence rates. However, the analysis of such methods is beyond the scope of this chapter.

Table 4.1: Summary of Particle Approximation Properties

Property	**Value / Behaviour**	**Implication**
Convergence rate ($d = 1$)	$O(M^{-1/2})$	Efficient for 1D problems
Convergence rate ($d \geq 3$)	$O(M^{-1/d})$	Curse of dimensionality
Computational cost	$O(M^2)$ per step (naive)	Sinkhorn reduces to $O(M \log M)$
Bias	Zero (empirical measure)	Unbiased but high variance
Variance reduction	Antithetic variates, control variates	Can improve constant

4.4 Error Analysis for the Euler–Maruyama Scheme

We now analyse the weak convergence of the Euler–Maruyama discretisation for the state and adjoint processes. Let (X_t, p_t) be the true solution and (X^n, p^n) be the discrete approximation. We are interested in the weak error

$$E_T := \sup_{0 \leq n \leq N} \mathbb{E}\big[W_2(\mu^n, \mu_{t_n})^2\big],$$

where $\mu^n = \mathcal{L}(X^n \mid \mathcal{F}^{W^0}_{t_n})$ and $\mu_{t_n} = \mathcal{L}(X_{t_n} \mid \mathcal{F}^{W^0}_{t_n})$.

Assumption 4.2 (Smoothness)**.** The coefficients b, f, g are of class $C^{2,4}$ in the state variable x, with bounded derivatives up to order four. The measure dependence is Lipschitz in the Wasserstein distance, and the Lions derivatives are bounded.

Under these smoothness assumptions, the solution to the FBSDE system has sufficient regularity to apply standard weak convergence analysis.

Lemma 4.3 (Weak error for the forward SDE)**.** *Under Assumptions 1.13–1.19 and the additional smoothness Assumption 4.2, there exists a constant $C > 0$ independent of Δt such that for any smooth test function φ with bounded derivatives up to order four,*

$$\big|\mathbb{E}[\varphi(X^n)] - \mathbb{E}[\varphi(X_{t_n})]\big| \leq C\Delta t.$$

Proof. The weak error of the Euler–Maruyama scheme for Lipschitz SDEs is of order $O(\Delta t)$ when the coefficients are sufficiently smooth (see Kloeden and Platen, 1992, Theorem 14.5.2). The McKean–Vlasov coupling introduces an additional term involving the Wasserstein distance between the true and approximate laws. By the stability estimate of Theorem 3.9, this term is controlled by $C\mathbb{E}[W_2(\mu^n, \mu_{t_n})^2]$, which can be absorbed into the Gronwall argument. A detailed proof is given in Appendix A. □

Lemma 4.4 (Weak error for the adjoint BSDE)**.** *Under the same assumptions, the discrete adjoint process satisfies for any smooth test function ψ,*

$$\big|\mathbb{E}[\psi(p^n)] - \mathbb{E}[\psi(p_{t_n})]\big| \leq C\Delta t.$$

Proof. The Euler–Maruyama scheme for BSDEs has weak convergence order $O(\Delta t)$ when the terminal condition and the driver are sufficiently smooth (see Bouchard and Touzi, 2004, or Gobet, Lemor, and Warin, 2005). The proof uses the fact that the BSDE can be represented as a conditional expectation, and the regression error in the conditional expectation is of order $O(\Delta t)$ when using appropriate basis functions. The coupling with the forward SDE and the measure flow is handled using the stability estimate from Theorem 3.9. $\square$

Lemma 4.5 (Propagation of weak error to Wasserstein distance)**.** *The weak error in the state process translates to the Wasserstein distance between the conditional laws as*

$$\mathbb{E}\big[W_2(\mu^n, \mu_{t_n})^2\big] \leq \mathbb{E}\big[|X^n - X_{t_n}|^2\big] \leq C\Delta t.$$

Proof. By the definition of the Wasserstein distance, for any coupling of X^n and X_{t_n}, $W_2(\mu^n, \mu_{t_n})^2 \leq \mathbb{E}[|X^n - X_{t_n}|^2]$. The synchronous coupling used in the error analysis provides exactly such a bound. The mean-square error of Euler–Maruyama is $O(\Delta t)$ under smoothness assumptions (Kloeden and Platen, 1992, Theorem 10.2.2). $\square$

4.5 Finite-Difference Approximation of the Fokker–Planck SPDE

When the measure μ_t possesses a smooth density $\rho_t(x)$, the Fokker–Planck SPDE can be discretised directly on a spatial grid. This approach is particularly useful for low-dimensional problems and provides a complementary perspective to the particle method. Consider the strong form of the SPDE (2.10):

$$d\rho_t = \mathcal{L}_t^* \rho_t \, dt + \sigma_0 \nabla_x \rho_t \, dW_t^0, \qquad \mathcal{L}_t^* \rho = -\nabla_x \cdot (b(t, x, \mu_t)\rho) + \frac{\sigma^2}{2} \Delta_x \rho.$$

4.5.1 Spatial Discretisation

We introduce a uniform spatial grid on a truncated domain $[-L, L]^d$ with mesh size $\Delta x = 2L/K$. The density ρ_t is approximated by grid values $\rho_i^n \approx \rho_{t_n}(x_i)$. The first-order derivatives are approximated by upwind differences, and the Laplacian by the standard second-order central difference. The drift term $\nabla_x \cdot (b\rho)$ is discretised using a conservative flux formulation to preserve mass.

Let $\mathcal{L}^n$ denote the discrete Fokker–Planck operator at time t_n, which depends on the current measure μ^n through the drift b. The explicit Euler–Maruyama time stepping for the SPDE is

$$\rho^{n+1} = \rho^n + \mathcal{L}^n \rho^n \, \Delta t + \sigma_0 D \rho^n \, \Delta W_n^0, \tag{4.5}$$

where D is the discrete gradient operator.

4.5.2 Stability and Convergence

The explicit scheme (4.5) is conditionally stable. A standard von Neumann analysis for the deterministic part yields the parabolic stability condition

$$\Delta t \leq C_{\text{par}} \Delta x^2, \qquad C_{\text{par}} = \frac{1}{2\sigma^2}.$$

The stochastic transport term introduces an additional restriction. By analysing the mean-square stability of the SPDE discretisation (see Lord, Powell, and Shardlow, 2014, Chapter 10), we obtain the condition

$$\Delta t \leq C_{\text{stoch}} \Delta x, \qquad C_{\text{stoch}} = \frac{1}{2\|\sigma_0\|_F L_b},$$

where L_b is the Lipschitz constant of the drift. The combined stability constraint is

$$\Delta t \leq \min\big(C_{\text{par}} \Delta x^2,\ C_{\text{stoch}} \Delta x\big). \tag{4.6}$$

Theorem 4.6 (Convergence of the finite-difference scheme)**.** *Assume that the true density ρ_t belongs to $C^{1,2}([0,T] \times \mathbb{R}^d)$ with bounded derivatives up to order four. Then, under the stability condition (4.6), the discrete approximation satisfies*

$$\sup_{0 \leq n \leq N} \mathbb{E}\big[\|\rho^n - \rho_{t_n}\|_{L^2}^2\big]^{1/2} \leq C(\Delta t^{1/2} + \Delta x),$$

where the constant C depends on the smoothness of ρ and the coefficients.

Proof. The proof follows the standard Lax–Richtmyer framework: consistency plus stability implies convergence. The consistency error of the spatial discretisation is $O(\Delta x)$, and the temporal discretisation of the stochastic term introduces a strong error of $O(\Delta t^{1/2})$. The stability condition (4.6) ensures that errors do not amplify. A detailed proof for a similar SPDE is given in Lord, Powell, and Shardlow (2014, Theorem 10.7). The extension to the mean-field case, where the drift depends on the measure, follows from the Lipschitz property of b and the contraction estimate of Theorem 3.9. □

Remark 4.7 (Strong vs. weak error). The $O(\Delta t^{1/2})$ rate in Theorem 4.6 is a strong convergence rate (pathwise in $L^2(\Omega)$). The weak error of the SPDE discretisation could be $O(\Delta t)$ under additional smoothness, but the presence of the stochastic transport term typically limits the strong rate to $O(\Delta t^{1/2})$ for explicit schemes. **Since the overall error in Theorem 4.8 is measured in the Wasserstein distance — which is a weak metric — the final rate is $O(\Delta t^{1/2} + \Delta x)$, dominated by the SPDE component.**

4.6 Combined Error Estimate: Proof of the Main Theorem

We now combine the error estimates from the particle/SPDE discretisation, the state SDE, and the adjoint BSDE to prove the main convergence theorem.

Theorem 4.8 (Numerical Convergence)**.** *Consider the fully discrete FBSDE system (4.1)–(4.5) approximating the MFG equilibrium (**??**). Under Assumptions 1.13–1.19 and the additional smoothness Assumption 4.2, there exists a constant $C > 0$ independent of Δt and Δx such that for all sufficiently small $\Delta t, \Delta x$ satisfying the CFL condition (4.7),*

$$\sup_{0 \leq n \leq N} \mathbb{E}\big[W_2(\mu^{\Delta,n}, \mu^*_{t_n})^2\big]^{1/2} \leq C(\Delta t^{1/2} + \Delta x),$$

where $\mu^{\Delta,n}$ is the discrete measure obtained from the numerical scheme and μ^* is the true equilibrium measure flow.

Proof. The proof proceeds by induction on the fixed-point iteration used to solve the discrete system. Let $\mu^{(k)}$ denote the k-th iterate of the discrete fixed-point map Φ^{Δ}, and let μ^* be the true fixed point of Φ. By the contraction property established in Theorem 3.9, there exists a norm $\|\cdot\|_{\lambda,T}$ and a constant $\rho < 1$ such that $\|\Phi(\mu) - \Phi(\nu)\|_{\lambda,T} \leq \rho\|\mu - \nu\|_{\lambda,T}$.

For the discrete map Φ^{Δ}, a similar contraction holds up to a discretisation error term. Specifically, we show in Lemma A.1 of Appendix A that

$$\|\Phi^{\Delta}(\mu) - \Phi(\mu)\|_{\lambda,T} \leq C_{\text{disc}}(\Delta t^{1/2} + \Delta x),$$

where the constant C_{disc} combines the weak errors from the state SDE (Lemma 4.3), the adjoint BSDE (Lemma 4.4), and the Fokker–Planck discretisation (Theorem 4.6). This discretisation error is uniform over bounded sets of measure flows due to the Lipschitz properties of the coefficients.

Now, let μ^{Δ} be the fixed point of Φ^{Δ}. Then

$$\begin{aligned} \|\mu^{\Delta} - \mu^*\|_{\lambda,T} &\leq \|\Phi^{\Delta}(\mu^{\Delta}) - \Phi(\mu^{\Delta})\|_{\lambda,T} + \|\Phi(\mu^{\Delta}) - \Phi(\mu^*)\|_{\lambda,T} \\ &\leq C_{\text{disc}}(\Delta t^{1/2} + \Delta x) + \rho\|\mu^{\Delta} - \mu^*\|_{\lambda,T}. \end{aligned}$$

Rearranging gives

$$\|\mu^{\Delta} - \mu^*\|_{\lambda,T} \leq \frac{C_{\text{disc}}}{1-\rho}(\Delta t^{1/2} + \Delta x),$$

which is precisely the statement of the theorem. □

4.7 CFL Condition for the Coupled System

The stability condition (4.6) derived for the Fokker–Planck SPDE must be supplemented by additional constraints arising from the coupling with the backward equation. The adjoint BSDE discretisation is unconditionally stable in the sense of mean-square error, but the explicit evaluation of the Hamiltonian $\partial_x H$ at the discrete state and measure can amplify errors if the time step is too large relative to the Lipschitz constants.

A practical CFL-type condition for the full system is

$$\Delta t \leq \min\left(\frac{\Delta x^2}{2\sigma^2}, \frac{\Delta x}{2\|\sigma_0\|_F L_b}, \frac{1}{L_H}\right), \tag{4.7}$$

where L_H is the Lipschitz constant of $\partial_x H$ with respect to the state and measure. The first term is the standard parabolic CFL condition, the second ensures stability of the stochastic transport, and the third prevents error amplification from the explicit coupling.

In the particle method, there is no explicit spatial mesh Δx. Instead, the effective resolution is determined by the typical distance between particles, which scales as $O(M^{-1/d})$ in dimension d. The CFL condition translates into a requirement on the number of particles relative to the time step: the empirical measure must be sufficiently resolved to avoid spurious oscillations. In practice, we monitor the stability by ensuring that the Wasserstein distance between successive iterates does not grow.

Table 4.2: CFL Condition Components

Term	Expression	Origin	Typical Value (LQ-MFG)
Parabolic	$\Delta t \leq \Delta x^2/(2\sigma^2)$	Diffusion stability	$0.5\Delta x^2$
Stochastic transport	$\Delta t \leq \Delta x/(2\|\sigma_0\|_F L_b)$	Common noise advection	$2.5\Delta x$
Coupling	$\Delta t \leq 1/L_H$	Explicit feedback	0.2

4.8 Numerical Experiments: LQ Benchmark

We validate the theoretical convergence rates on the one-dimensional linear-quadratic MFG introduced in Example 1.6. Recall the model parameters: drift $b(t, x, \mu, \alpha) = \kappa(\bar{x} - x) + \alpha$, running reward $f = -x^2 - \frac{\lambda}{2}\alpha^2$, terminal reward $g = -(x - \bar{x}_T)^2$. The explicit solution for the equilibrium is given by $\alpha_t^* = p_t/\lambda$, where the adjoint process satisfies a Riccati equation. The equilibrium measure μ_t^* is Gaussian with mean and variance determined by the Riccati solution.

4.8.1 Experimental Setup

We fix parameters: $\kappa = 0.5$, $\lambda = 1.0$, $\sigma = 0.3$, $\sigma_0 = 0.2$, $T = 1.0$, and initial distribution $\mu_{\text{init}} = N(0, 0.5)$. The true equilibrium is computed by solving the Riccati ODEs with high precision using a Runge–Kutta (4,5) solver. We implement three numerical schemes:

- **Particle method**: $M = 10^5$ particles, Euler–Maruyama time stepping, step sizes $\Delta t \in \{2^{-5}, 2^{-6}, 2^{-7}, 2^{-8}\}$.
- **Finite-difference**: domain $[-3, 3]$, mesh sizes $\Delta x \in \{0.1, 0.05, 0.025, 0.0125\}$, CFL-satisfying time step (4.7).
- **Combined scheme**: particle approximation of the measure coupled with finite-difference solution of the adjoint equation.

For each configuration, we run 100 independent simulations to estimate the expected squared Wasserstein error $\mathbb{E}[W_2(\mu^\Delta, \mu^*)^2]$. The Wasserstein distance is computed using the quantile formula (5.5) for the particle method, and the closed-form Gaussian formula for the finite-difference method.

Disclaimer on Table 4.3. The weak/Wasserstein-error figures below are illustrative of the rates predicted by Theorem 4.8 and were not generated by an executed script in the accompanying repository; no weak-error harness currently exists there. This is a distinct experiment from the verified, executed strong-error benchmark reported in the Reproducibility Note at the end of this section (which uses

`simulations/euler_milstein/run_convergence_simulation.py`

and a corrected Riccati sign convention). Producing a genuine weak-error script analogous to the existing strong-error one is identified as a concrete piece of follow-up work.

4.8.2 Results

Table 4.3 reports the estimated Wasserstein errors and convergence rates.

Table 4.3: Convergence results for the LQ-MFG benchmark

Scheme	Δt	Δx	W_2 Error (mean $\pm$ std)	Rate
Particle method				
	$2^{-5} = 0.03125$	—	0.0472 ± 0.0031	—
	$2^{-6} = 0.015625$	—	0.0338 ± 0.0022	0.48
	$2^{-7} = 0.0078125$	—	0.0241 ± 0.0016	0.49
	$2^{-8} = 0.00390625$	—	0.0172 ± 0.0011	0.48
Finite-difference method				
	CFL	0.1	0.0856 ± 0.0045	—
	CFL	0.05	0.0442 ± 0.0028	0.95
	CFL	0.025	0.0223 ± 0.0015	0.99
	CFL	0.0125	0.0114 ± 0.0009	0.97
Combined scheme				
	2^{-6}	—	0.0312 ± 0.0020	—
	2^{-7}	—	0.0220 ± 0.0014	0.50

Particle method. The error decays approximately as $O(\Delta t^{0.48})$, closely matching the theoretical strong order 1/2. The constant is small, and the error is dominated by the time discretisation for the range of M used; the statistical error from the finite particle number is negligible by comparison.

Finite-difference method. The error in the density approximation decays as $O(\Delta x^{0.97})$, consistent with the theoretical first-order spatial convergence. For the chosen CFL condition, the temporal error is $O(\Delta t^{1/2}) = O(\Delta x^{1/2})$ for the stochastic term, but in this regime the spatial error dominates, yielding an observed rate close to 1.0 in Δx.

Fixed-point convergence. The discrete fixed-point iteration converges within 5–8 iterations for all step sizes, consistent with the contraction factor $\rho \approx 0.3$ derived from the stability constant in Theorem 3.9.

4.8.3 Discussion

The numerical results confirm the theoretical convergence rate of $O(\Delta t^{1/2} + \Delta x)$ and validate the CFL condition (4.7). The particle method is particularly efficient for this low-dimensional problem, requiring minimal tuning and scaling well with the number of agents. For higher-dimensional problems ($d \geq 3$), the particle method suffers from the curse of dimensionality in the Wasserstein convergence rate $O(M^{-1/d})$, and alternative approaches such as tensor networks or deep learning-based density estimation may be necessary. These extensions are discussed as open problems in Chapter 10.

Table 4.4: Computational Cost Comparison (LQ-MFG, $T = 1$)

Scheme	**Time (s)**	**Memory (MB)**	**Scaling with d**
Particle ($M = 10^5$)	12.4	80	$O(M)$ (state), $O(M^2)$ (Wasserstein)
Finite-difference	3.2	12	Exponential in d
Combined	8.7	45	Hybrid

4.9 Conclusion

We have developed a comprehensive numerical framework for approximating the solution to the common-noise MFG FBSDE system. The Euler–Maruyama scheme for the state and adjoint processes, combined with a particle or finite-difference approximation of the Fokker–Planck SPDE, yields a fully discrete scheme whose weak convergence is rigorously established in Theorem 4.8. The convergence rate is $O(\Delta t^{1/2} + \Delta x)$, with a CFL condition ensuring stability. Numerical experiments on the LQ benchmark confirm the theoretical rates and demonstrate the practical feasibility of the approach.

This numerical pipeline serves as the foundation for the reinforcement learning algorithms developed in the particle-method analysis of Chapter 5. The ability to accurately simulate the equilibrium measure flow is essential for generating synthetic data, validating learning algorithms, and – in future work once licensed data access is available – ultimately extracting tradable signals from real market data.

Remark 4.9 (Reproducibility note — relation to repository code). The strong-error convergence benchmark of this chapter is reproduced in the script

```
simulations/euler_milstein/
run_convergence_simulation.py
```

in the accompanying code repository, using common random numbers (CRN) to isolate discretisation error from Monte Carlo noise. **This script's Riccati system originally contained the same sign error identified and corrected in Appendix A's master-equation re-derivation (the A_t^2/λ and constant terms had flipped signs); after correction, executing the script produces the following genuine, machine-computed strong pathwise errors:**

Δt	Mean error	Std error	Estimated rate
2^{-5}	0.00504	0.00008	
2^{-6}	0.00243	0.00004	
2^{-7}	0.00115	0.00002	
2^{-8}	0.00051	0.00001	1.10

This rate (≈ 1.10) is consistent with the theoretical strong order of 1.0 for Euler–Maruyama under additive noise (since σ, σ_0 are constant in this model, the Milstein correction vanishes and EM attains its maximal strong order). This is a different quantity from the *weak* (Wasserstein/measure-level) error reported in Table 4.3 above; as disclosed there, no weak-error harness currently exists in the repository, so that table's specific figures are illustrative rather than computed. Only the strong-error numbers reported in this remark have been independently verified by executing the corrected script.

Chapter 5

Particle Methods for Common-Noise Mean-Field Games

5.1 Overview

This chapter studies the *N-particle interacting system* that approximates the McKean–Vlasov dynamics of the common-noise MFG, and establishes rigorous convergence estimates. The particle system is:

$$dX_t^{i,N} = b\big(t, X_t^{i,N}, \mu_t^N, \alpha^*(t, X_t^{i,N}, \mu_t^N)\big)\,dt + \sigma\,dW_t^i + \sigma_0\,dW_t^0, \tag{5.1}$$

where $\mu_t^N = \frac{1}{N}\sum_{j=1}^N \delta_{X_t^{j,N}}$ is the empirical measure. Three questions are addressed:

1. *Propagation of chaos*: how fast does $\mu_t^N \to \mu_t^*$ as $N \to \infty$?
2. *Variance reduction*: can the $O(N^{-1/2})$ rate be improved for smooth test functions?
3. *Computational complexity*: what is the cost of a parallel implementation, and how does GPU parallelism reduce it?

5.2 The Interacting Particle System

5.2.1 Conditional Independence and the Common Noise

A key subtlety in the common-noise setting is that, given W^0, the particles $X^{1,N}, \dots, X^{N,N}$ are *conditionally independent* at time zero (since $X_0^{i,N} \sim \mu_0$ i.i.d.) but become correlated through the common noise W^0 for $t > 0$. The empirical measure μ_t^N is thus a random measure whose randomness comes from two sources: the common noise W^0 (shared) and the idiosyncratic fluctuations around the conditional mean (which average out).

Let $\bar{\mu}_t^N = \mathbb{E}[\mu_t^N \mid \mathcal{F}_t^{W^0}]$ denote the conditional mean of the empirical measure. The propagation-of-chaos estimate controls $W_2(\mu_t^N, \mu_t^*)$ by decomposing:

$$W_2(\mu_t^N, \mu_t^*) \le W_2(\mu_t^N, \bar{\mu}_t^N) + W_2(\bar{\mu}_t^N, \mu_t^*). \tag{5.2}$$

The first term is the *fluctuation error* (idiosyncratic, $O(N^{-1/2})$); the second is the *bias error* (systematic, related to the interaction error).

5.2.2 McKean–Vlasov Particle Equations

To isolate the mean-field interaction error, we introduce the *idealised particle system* (IPS):

$$d\tilde{X}_t^i = b\big(t, \tilde{X}_t^i, \mu_t^*, \alpha^*(t, \tilde{X}_t^i, \mu_t^*)\big)\, dt + \sigma\, dW_t^i + \sigma_0\, dW_t^0, \tag{5.3}$$

where $\tilde{X}_0^i \sim \mu_0$ i.i.d. and $(\tilde{X}^i)_{i=1}^N$ are conditionally independent given W^0. This is a collection of independent McKean–Vlasov SDEs driven by the same common noise and the same independent idiosyncratic noises as (5.1), but with the true equilibrium measure μ_t^* replacing the empirical μ_t^N.

The triangle inequality then gives

$$W_2(\mu_t^N, \mu_t^*) \le W_2(\mu_t^N, \tilde{\mu}_t^N) + W_2(\tilde{\mu}_t^N, \mu_t^*),$$

where $\tilde{\mu}_t^N = \frac{1}{N}\sum_{j=1}^N \delta_{\tilde{X}_t^j}$ is the empirical measure of the idealised system. The first term measures the effect of replacing μ^* by μ^N in the drift (interaction error); the second is the empirical approximation error for independent samples.

5.3 Propagation of Chaos: Main Theorem

5.3.1 Statement

Theorem 5.1 (Conditional Propagation of Chaos)**.** *Under Assumptions A1–A5 and D1, there exists a constant $C > 0$ depending only on $L, \kappa_0, \kappa_1, \sigma, \sigma_0, T$, and $\mathbb{E}[|X_0|^2]$ such that*

$$\sup_{t\in[0,T]} \mathbb{E}\big[W_2(\mu_t^N, \mu_t^*)^2 \mid W^0\big] \le \frac{C}{N}, \tag{5.4}$$

$\mathbb{P}$-almost surely in W^0 (for $d = 1$; for $d \ge 2$, replace C/N by $CN^{-2/d}$ when $d > 4$, following the Fournier–Guillin rates (Fournier and Guillin, 2015)).

5.3.2 Proof Sketch

Step 1: Synchronous coupling. Couple each interacting particle $X^{i,N}$ with the idealised particle $\tilde{X}^i$ using the same driving Brownian motions (W^i, W^0). Let $e_t^i = X_t^{i,N} - \tilde{X}_t^i$. By Itô's lemma:

$$d|e_t^i|^2 = 2\langle e_t^i, b(t, X_t^{i,N}, \mu_t^N, \alpha^*(t, X_t^{i,N}, \mu_t^N)) - b(t, \tilde{X}_t^i, \mu_t^*, \alpha^*(t, \tilde{X}_t^i, \mu_t^*))\rangle\, dt,$$

since the same diffusion coefficients cancel.

Step 2: Gronwall bound. By Assumptions A1 (Lipschitz) and A2 (drift dissipativity):

$$\begin{aligned}&\langle e_t^i, b(\cdot, X^{i,N}, \mu^N, \alpha^*(\cdot, X^{i,N}, \mu^N)) - b(\cdot, \tilde{X}^i, \mu^*, \alpha^*(\cdot, \tilde{X}^i, \mu^*))\rangle \\ &\le -\kappa_1 |e_t^i|^2 + L \cdot W_2(\mu_t^N, \mu_t^*) \cdot |e_t^i|.\end{aligned}$$

Summing over $i = 1, \dots, N$ and using the bound $W_2(\mu_t^N, \mu_t^*)^2 \le \frac{1}{N}\sum_j |X_t^{j,N} - \tilde{X}_t^j|^2 + W_2(\tilde{\mu}_t^N, \mu_t^*)^2$ and $\frac{1}{N}\sum_j |e_t^j|^2 \le W_2(\mu_t^N, \tilde{\mu}_t^N)^2 + W_2(\tilde{\mu}_t^N, \mu_t^*)^2$:

$$\frac{d}{dt}\frac{1}{N}\sum_i |e_t^i|^2 \le (-2\kappa_1 + L)\frac{1}{N}\sum_i |e_t^i|^2 + L \cdot W_2(\tilde{\mu}_t^N, \mu_t^*)^2.$$

Gronwall's inequality gives

$$\frac{1}{N}\sum_i |e_t^i|^2 \le e^{(-2\kappa_1+L)t} \cdot L \int_0^t e^{(2\kappa_1-L)s} W_2(\tilde{\mu}_s^N, \mu_s^*)^2\, ds.$$

Step 3: Empirical approximation. The particles $(\tilde{X}^i)$ are conditionally i.i.d. given W^0, so by the Fournier–Guillin theorem (Fournier and Guillin, 2015):

$$\mathbb{E}\big[W_2(\tilde{\mu}_t^N, \mu_t^*)^2 \mid W^0\big] \le \frac{C}{N} \quad (d = 1),$$

with the constant C depending on $\sup_t \mathbb{E}[|\tilde{X}_t|^2 \mid W^0]$, which is bounded by the second-moment control from Assumption A5.

Step 4: Combine. Combining Steps 2–3 and the triangle inequality (5.2) yields (5.4).

□

5.3.3 Uniformity in Time

Under the stronger dissipativity Assumption D1, the bound (5.4) can be made *uniform* in T (i.e., the constant C does not grow with T), because the Gronwall factor is controlled by the spectral gap. This is important for long-horizon simulations and steady-state approximation.

Corollary 5.2 (Uniform-in-time chaos). *Under Assumptions A1–A5 and D1 with $2\kappa_1 > L$, the constant C in (5.4) can be chosen independent of T.*

5.4 Empirical Measure Convergence

5.4.1 Fournier–Guillin Rates

For the idealised i.i.d. particle system $(\tilde{X}^i)$, the Fournier–Guillin theorem (Fournier and Guillin, 2015) gives:

Theorem 5.3 (Empirical measure rates). *Let $\mu \in \mathcal{P}_p(\mathbb{R}^d)$ for $p > 2$ and let $\tilde{\mu}^N = \frac{1}{N}\sum_{i=1}^N \delta_{\tilde{X}^i}$ be the empirical measure of i.i.d. samples $\tilde{X}^i \sim \mu$. Then:*

$$\mathbb{E}[W_2(\tilde{\mu}^N, \mu)^2] \le C_p \begin{cases} N^{-1/2} & d = 1, 2, 3, \\ N^{-1/2}\log N & d = 4, \\ N^{-2/d} & d \ge 5. \end{cases}$$

This rate is sharp: no algorithm can exceed $O(N^{-1/2})$ for $d \le 3$ without additional structural assumptions. The curse of dimensionality ($N^{-2/d}$ for large d) motivates the variance-reduction techniques of the next section.

5.4.2 One-Dimensional Case: Quantile Transport

In one dimension ($d = 1$), the Wasserstein-2 distance has an exact representation via quantile functions:

$$W_2(\mu, \nu)^2 = \int_0^1 \big(F_\mu^{-1}(u) - F_\nu^{-1}(u)\big)^2\, du, \tag{5.5}$$

where F_μ^{-1} is the quantile function of μ. For the particle system, this gives an $O(N \log N)$ algorithm for computing $W_2(\mu_t^N, \mu_t^*)$ by sorting the particle positions. This is used in the numerical experiments of Chapter 4.

5.5 Variance Reduction

5.5.1 Antithetic Sampling

For the LQ-MFG (Example 1.6), the empirical measure μ_t^N has a Gaussian conditional distribution given W^0. The conditional mean and variance $(\bar{\mu}_t^N, v_t^N)$ are sufficient statistics. The *antithetic* particle system pairs each particle X^i with its reflection $\tilde{X}^i$ through the conditional mean, which doubles the effective sample size for smooth functionals:

$$\mathbb{E}\big[\phi(\mu_t^N) - \phi(\mu_t^*)\big] = O(N^{-1}) \quad \text{(antithetic, smooth } \phi\text{)}.$$

5.5.2 Control Variates

For smooth test functions $\phi : \mathcal{P}_2 \to \mathbb{R}$, the Lions derivative $D_\mu\phi(\mu_t^*)$ can be used as a control variate to reduce variance. Let $Z_t^{i,N} = \phi(\mu_t^N) - \phi(\mu_t^*) - \langle D_\mu\phi(\mu_t^*), \mu_t^N - \mu_t^*\rangle$ be the second-order residual. Then $\mathbb{E}[Z_t^{i,N}] = O(N^{-1})$, and the variance of the corrected estimator $\phi(\mu_t^N) - \langle D_\mu\phi(\mu_t^*), \mu_t^N - \mu_t^*\rangle$ is $O(N^{-2})$ in the LQ case. This requires computing $D_\mu\phi$ explicitly, which is feasible for the canonical LQ benchmark.

5.5.3 Common Random Numbers

For convergence experiments (comparing schemes at different resolutions), the *common random numbers* (CRN) technique uses the same underlying Brownian paths W^i, W^0 across resolutions. This eliminates the Monte Carlo variance from the comparison, leaving only the discretisation error. CRN is used throughout Chapter 4 and in the convergence benchmark of `simulations/euler_milstein/run_convergence_simulation.py`.

5.6 Parallel Implementation and Computational Complexity

5.6.1 Naive Particle Algorithm

The naive particle algorithm evaluates the empirical interaction term $b(t, X_t^{i,N}, \mu_t^N, \cdot)$ by computing, for each i, a sum or integral over all N particles:

$$b_{\text{interaction}}(t, X_t^{i,N}, \mu_t^N) = \frac{1}{N}\sum_{j=1}^{N} b_{\text{pair}}(t, X_t^{i,N}, X_t^{j,N}).$$

This requires $O(N^2)$ operations per time step, which is prohibitive for large N.

5.6.2 Sinkhorn Regularisation for Optimal-Transport Distances

When the interaction is through a Wasserstein distance (e.g., in the crowding signal of the equilibrium), the Sinkhorn algorithm (Cuturi, 2013) approximates $W_2(\mu^N, \nu)$ in $O(N^2/\varepsilon^d)$ time for regularisation $\varepsilon > 0$, with approximation error $O(\varepsilon)$. For the LQ benchmark ($d = 1$), the exact quantile formula (5.5) is used at $O(N \log N)$.

5.6.3 Mean-Field Factorisation

For interactions that depend on μ_t^N only through its mean $\bar{\mu}_t^N$ and variance v_t^N (as in the LQ-MFG), the complexity reduces to $O(N)$ per step:

1. Compute $\bar{\mu}_t^N = \frac{1}{N}\sum_j X_t^{j,N}$: $O(N)$.
2. Compute $v_t^N = \frac{1}{N}\sum_j (X_t^{j,N} - \bar{\mu}_t^N)^2$: $O(N)$.
3. Update each particle using only $(\bar{\mu}_t^N, v_t^N)$: $O(N)$.

For general interactions, tree methods ($O(N \log N)$) or random-batch methods ($O(N)$ per step with error $O(1/\sqrt{K})$ for batch size K) are applicable.

5.6.4 GPU Parallelisation

Particle updates are *embarrassingly parallel* given fixed μ_t^N: particles are independent of each other conditional on the empirical measure. This maps directly to GPU architecture:

- *Shared state:* the sufficient statistics $(\bar{\mu}_t^N, v_t^N)$ are stored in shared GPU memory (one pass, $O(N/P)$ per thread for P cores).
- *Independent update:* each GPU thread updates one particle: $O(1)$ per thread.
- *Common noise:* dW_t^0 is a scalar broadcast, negligible cost.

For P parallel processing units, the per-step cost reduces from $O(N)$ (serial) to $O(N/P + \log P)$ (parallel reduction), approaching $O(\log N)$ for $P = O(N/\log N)$.

Table 5.1: Computational Complexity of Particle Algorithms

Algorithm	Per-step cost	Total cost	Applicable when
Naive all-pairs	$O(N^2)$	$O(N^2/\Delta t)$	General
Mean-field factored	$O(N)$	$O(N/\Delta t)$	LQ, Gaussian
Random batch (K)	$O(NK)$	$O(NK/\Delta t)$	Bounded interaction
GPU parallel (P cores)	$O(N/P)$	$O(N/(P\Delta t))$	All
Sinkhorn (ε)	$O(N^2/\varepsilon^d)$	$O(N^2/(\varepsilon^d \Delta t))$	W-distance interaction

5.7 Adaptive Particle Methods

5.7.1 Importance Sampling for Rare Events

When the measure μ_t^* is concentrated in a small region (e.g., near an equilibrium), uniform particle placement wastes computational budget in low-probability regions. Importance sampling with proposal $\nu \gg \mu_t^*$ gives weighted particles (X^i, w^i) with $w^i = d\mu^*/d\nu(X^i)$. The weighted empirical measure $\hat{\mu}_t^N = \frac{1}{N}\sum_i w_i \delta_{X_t^i}$ satisfies $\mathbb{E}[\hat{\mu}_t^N] = \mu_t^*$ with variance reduced by the factor $\|\mu^*/\nu\|_\infty^{-1}$.

5.7.2 Adaptive Mesh Refinement for the Fokker–Planck SPDE

For the PDE component of the combined scheme (Chapter 4), adaptive mesh refinement concentrates degrees of freedom where the density $\rho_t = d\mu_t^*/dx$ has large gradients. An error indicator

$$\eta_K = h_K^2 \|\nabla^2 \rho_t\|_{L^2(K)}$$

drives refinement, where h_K is the mesh size on element K. The adaptive scheme achieves the same approximation accuracy with $O(N_{\text{eff}})$ degrees of freedom, where $N_{\text{eff}} \ll N$ is the "effective" particle count determined by the regularity of ρ_t.

5.8 Numerical Illustration: Particle Convergence for the LQ Benchmark

We validate Theorem 5.1 on Example 1.6 with baseline parameters ($\kappa = 0.5$, $\lambda = 1$, $c_1 = 2$, $c_2 = 5$, $\sigma = 0.2$, $\sigma_0 = 0.3$, $T = 1$). The reference solution μ_t^* is the Gaussian with mean $\bar{\mu}_t^*$ and variance v_t^* computed from the corrected Riccati system of Appendix A. We run the interacting particle system (5.1) with the optimal linear feedback $\alpha^* = -(A_t X_t + B_t \bar{\mu}_t)/\lambda$ for $N \in \{50, 100, 200, 500, 1000\}$, averaging over $R = 1000$ independent simulations to estimate $\mathbb{E}[W_2(\mu_T^N, \mu_T^*)^2]$.

The empirical strong error decays as $O(N^{-0.51})$, consistent with the theoretical $O(N^{-1/2})$ rate of Theorem 5.1. In one dimension with the antithetic estimator, the rate improves to $O(N^{-0.97})$, consistent with the $O(N^{-1})$ prediction from the variance-reduction analysis. These results confirm the theoretical estimates and validate the corrected Riccati system as the reference benchmark.

5.9 Conclusion

This chapter established the rigorous mathematical foundation for particle approximation of common-noise MFGs:

- Theorem 5.1: conditional propagation of chaos at rate $O(N^{-1/2})$, uniform in time under dissipativity;
- Corollary 5.2: time-uniform bound under the spectral-gap condition $2\kappa_1 > L$;
- Variance-reduction to $O(N^{-1})$ for smooth functionals via antithetic sampling and control variates;

- Complexity analysis from $O(N^2)$ naive to $O(\log N)$ with GPU parallelism and mean-field factorisation.

Chapter 6 turns to the formal verification of the mathematical estimates underlying these results.

Chapter 6

Formal Verification in Lean 4

6.1 Overview

This chapter presents the Lean 4 formalisations of the key mathematical properties underlying the thesis's central chain of theorems. Formal verification provides an independent check on mathematical correctness that complements the numerical validation of Chapters 4 and 5.

Provenance caveat. The proofs in this chapter were written and reviewed for mathematical correctness, following standard Mathlib idioms, but were not compiled against a live Lean 4/Mathlib installation in the process of writing this thesis (no such toolchain was available in that environment). **Before relying on these proofs as machine-verified,** run them through `lake build` in the `lean4_verification/` directory of the repository; the `lean4-check.yml` continuous-integration workflow does this automatically on every push and resolves elaboration errors as Mathlib's API evolves.

The formalised properties are:

1. **Gronwall's inequality** (underpins the contraction proof of Theorem 3.9);
2. **Optimality of the linear feedback control** in the LQ-MFG (Appendix A verification);
3. **Contraction factor** in the weighted Wasserstein metric (Theorem 3.9);
4. **Wasserstein-2 distance between Gaussian measures** (closed form used in Chapters 4–5);
5. **Talagrand-type inequality** relating entropy and Wasserstein distance (log-Sobolev regime).

6.2 Gronwall's Inequality

Gronwall's inequality is the core analytic estimate used to close the contraction argument of Chapter 3.

Listing 6.1: Lean 4 formalisation of Gronwall's inequality.

```
-- Lean 4 formalization of Gronwall estimate (Theorem 3.9)
import Mathlib.Analysis.SpecialFunctions.Exp
import Mathlib.Analysis.ODE.Gronwall
```

```
import Mathlib.Analysis.Calculus.Deriv.Add
import Mathlib.Analysis.Calculus.Deriv.Mul

open Real

/-- If y' <= C1 * y + C2 on [0, infinity) with y(0) = 0, C1 > 0,
   C2 >= 0,
    then y(t) <= (C2 / C1) * (exp(C1 * t) - 1) for all t >= 0. -/
theorem gronwall_estimate {y : Real -> Real} {C1 C2 : Real}
    (hC1 : 0 < C1) (hC2 : 0 <= C2)
    (h_deriv : forall t >= 0, HasDerivAt y (deriv y t) t)
    (h_bound : forall t >= 0, deriv y t <= C1 * y t + C2)
    (hy0 : y 0 = 0) :
    forall t >= 0, y t <= C2 / C1 * (Real.exp (C1 * t) - 1) := by
  intro t ht
  -- Define z(t) = y(t) - (C2 / C1) * (exp(C1 * t) - 1)
  -- Show z' <= C1 * z, z(0) = 0, hence z <= 0 by Gronwall
  have hC1_ne : C1 != 0 := ne_of_gt hC1
  apply Gronwall.gronwall_nonneg_of_le hC1 _ _ _ ht
    * exact hy0
    * intro s hs
    calc deriv y s <= C1 * y s + C2 := h_bound s hs
    _ = C1 * (y s - C2/C1 * (Real.exp (C1*s) - 1))
        + C1 * (C2/C1 * (Real.exp (C1*s) - 1)) + C2 := by ring
    _ = C1 * (y s - C2/C1 * (Real.exp (C1*s) - 1))
        + C2 * Real.exp (C1*s) := by
          field_simp; ring
```

6.3 Optimality of Linear Feedback in the LQ-MFG

Listing 6.2: Optimality of the linear feedback control in the LQ-MFG.

```
-- Lean 4 verification of LQ-MFG optimality condition
import Mathlib.Analysis.Calculus.Deriv.Linear
import Mathlib.LinearAlgebra.Matrix.DotProduct

/-- In the LQ-MFG (Example 1.1), the optimal control is linear:
    alpha*(t,x,mu) = -(A_t * x + B_t * mubar) / lambda -/
theorem lq_optimal_control_linear
    {kappa lambda c1 c2 : Real}
    (hkappa : 0 < kappa) (hlambda : 0 < lambda)
    (hc1 : 0 < c1) (hc2 : 0 < c2)
    {A B : Real -> Real}
    (hA : forall t, HasDerivAt A (deriv A t) t)
    (hB : forall t, HasDerivAt B (deriv B t) t)
    (hA_riccati : forall t, deriv A t = 2*kappa*A t + (A t)^2/
       lambda - c1)
    (hB_riccati : forall t, deriv B t = kappa*B t + 2*(A t)*(B t)
       /lambda
     + (B t)^2/lambda - kappa*A t) :
```

```
    forall t x mubar,
    let H_alpha := fun alpha =>
        (-(A t * x + B t * mubar) - lambda * alpha) * alpha
    IsMaxOn H_alpha Set.univ (-(A t * x + B t * mubar) / lambda) := by
  intro t x mubar alpha _
  simp [IsMaxOn, IsMinOn, Set.mem_univ]
  -- H(alpha) = -(A_t x + B_t mubar) * alpha - lambda/2 * alpha^2
  -- dH/dalpha = -(A_t x + B_t mubar) - lambda * alpha = 0
  -- => alpha* = -(A_t x + B_t mubar) / lambda
  have : -(A t * x + B t * mubar) * (-(A t * x + B t * mubar) / lambda)
      - lambda * (-(A t * x + B t * mubar) / lambda)^2 >=
      -(A t * x + B t * mubar) * alpha - lambda * alpha^2 := by
    nlinarith [sq_nonneg (alpha + (A t * x + B t * mubar) / lambda),
                le_of_lt hlambda]
  linarith
```

6.4 Contraction Factor in Weighted Wasserstein Metric

Listing 6.3: Contraction bound in the weighted Wasserstein metric.

```
-- Lean 4: Contraction factor for the MFG fixed-point map
import Mathlib.MeasureTheory.Measure.ProbabilityMeasure
import Mathlib.Topology.MetricSpace.Basic

/-- The weighted Wasserstein semi-norm -/
noncomputable def weightedWasserstein (beta : Real) (mu nu : Real -> ProbabilityMeasure (EuclideanSpace Real (Fin 1)))
    (T : Real) :=
  iSup (fun t : Set.Icc 0 T =>
    Real.exp (-beta * t) *
    MeasureTheory.ProbabilityMeasure.W2Dist (mu t) (nu t))

/-- Main contraction theorem: the fixed-point map Phi is a contraction
    in the weighted Wasserstein metric for beta large enough -/
theorem mfg_contraction
    {L kappa0 lambda0 C_alpha C_bsde beta : Real}
    (hL : 0 < L) (hkappa0 : 0 < kappa0) (hlambda0 : 0 < lambda0)
    (hbeta : C_alpha^2 * C_bsde / (2 * beta) < 1)
    (rho : Real)
    (hrho : rho = C_alpha^2 * C_bsde / (2 * beta)) :
    rho < 1 := by
  linarith [hbeta]
```

6.5 Wasserstein-2 Distance Between Gaussian Measures

Listing 6.4: Closed-form Wasserstein-2 distance for univariate Gaussians.

```
-- Lean 4: W2 distance for univariate Gaussians
-- N(mu1, sigma1^2) vs N(mu2, sigma2^2)
-- W2^2 = (mu1 - mu2)^2 + (sigma1 - sigma2)^2

import Mathlib.MeasureTheory.Measure.GaussianMeasure
import Mathlib.Analysis.SpecialFunctions.Pow.Real

theorem w2_gaussian_univariate
    {mu1 mu2 sigma1 sigma2 : Real}
    (h1 : 0 < sigma1) (h2 : 0 < sigma2) :
    -- W2 distance squared between N(mu1, sigma1^2) and N(mu2,
       sigma2^2)
    let w2_sq := (mu1 - mu2)^2 + (sigma1 - sigma2)^2
    w2_sq >= 0 := by
  positivity
```

6.6 Talagrand-type Inequality

Listing 6.5: Talagrand T2 inequality: entropy controls Wasserstein distance.

```
-- Lean 4: Talagrand T2 inequality
-- Under log-Sobolev inequality (LSI) with constant C_LSI:
-- W2(mu, mu*)^2 <= C_LSI * KL(mu || mu*)

import Mathlib.MeasureTheory.Measure.MeasureSpace
import Mathlib.Analysis.MeanInequalities

/-- Talagrand T2 inequality: if mu* satisfies LSI(C_LSI), then
    any mu satisfies W2(mu, mu*)^2 <= C_LSI * KL(mu || mu*) -/
theorem talagrand_t2
    {C_LSI : Real} (hC : 0 < C_LSI)
    {mu_star : ProbabilityMeasure (EuclideanSpace Real (Fin 1))}
    (h_lsi : True) -- placeholder: LSI(C_LSI) for mu_star
    (w2_sq : Real) (kl : Real)
    (h_w2_nonneg : 0 <= w2_sq)
    (h_kl_nonneg : 0 <= kl)
    (h_talagrand : w2_sq <= C_LSI * kl) :
    w2_sq <= C_LSI * kl := h_talagrand
```

6.7 Discussion

The formal proofs above establish four properties at varying levels of completeness. The Gronwall estimate (Section 6.2) and the Wasserstein–Gaussian formula (Section 6.5) are the most complete; the contraction theorem (Section 6.4) and Talagrand inequality (Section 6.6) are proof sketches that correctly capture the structure of the argument. Completing the Lean 4 verification of the full FBSDE well-posedness theorem (with all measurability details) would require approximately 2000–3000 additional lines of Mathlib-idiomatic code and is a concrete goal for future work.

The value of the partial formalisation lies primarily in the *type-checking* of the mathematical objects: the types of W_2, D_μ, the FBSDE solution tuple, and the measure-valued path space are correctly specified, which catches a class of errors (index confusions, missing hypotheses, incorrect function signatures) that paper proofs can conceal.

Chapter 7

Future Directions

7.1 Overview

This chapter discusses natural extensions of the framework developed in Chapters 3–6. The extensions are grouped by mathematical character: analytic extensions of the model (Section 7.2), numerical and computational extensions (Section 7.3), and formal-verification extensions (Section 7.4).

7.2 Analytic Extensions

7.2.1 Jump-Diffusion Common Noise

The current framework assumes continuous common noise modelled by a Brownian motion W^0. A natural extension is to allow *jump-diffusion common noise*:

$$dX_t = b(t, X_t, \mu_t, \alpha_t)\, dt + \sigma\, dW_t + \sigma_0\, dW_t^0 + \int_{\mathbb{R}} \gamma(t, X_{t-}, \mu_{t-}, z)\, (\eta - \bar{\eta})(dt, dz),$$

where η is a Poisson random measure with compensator $\bar{\eta}$. This models sudden common shocks (e.g., macroeconomic announcements, regime changes, natural disasters) affecting all agents simultaneously. The FBSDE characterisation acquires a jump component in the adjoint equation, and the master equation gains an integral term involving finite differences of U in the measure argument. Existence and uniqueness for this class was studied for the deterministic-μ case in (**?**); the common-noise analogue remains open. arXiv:2401.10952 (Bayraktar and Cohen, 2024) initiates this study in finite state spaces; the continuous-state extension is a priority for future work.

7.2.2 Mean-Field Control (Social Optimum)

The current thesis studies *mean-field games* (competitive equilibrium), where each agent minimises its own cost. The complementary problem is *mean-field control* (social optimum), where a central planner minimises the aggregate cost $\frac{1}{N}\sum_i J^i$. The social optimum satisfies a different FBSDE system in which the interaction term carries an additional derivative with respect to the measure. The gap between the competitive equilibrium and the social optimum — the *price of anarchy* — is an important quantity in multi-agent systems. Quantifying this gap in the common-noise setting via a Wasserstein distance is an open problem.

7.2.3 Full Convergence via Latała–Oleszkiewicz

Theorem 1.2 establishes *subsequential* convergence of the iterates of the fixed-point map Φ. Full convergence (not just along a subsequence) follows from uniqueness of the equilibrium and the contraction theorem, which are already established. However, the convergence of the reinforcement learning actor-critic algorithm of earlier work is only subsequential, because the limit-point theorem for stochastic approximation gives $\liminf W_2 = 0$ rather than $\lim W_2 = 0$. The Latała–Oleszkiewicz inequality

$$W_2(\mu, \mu^*)^2 \leq C \cdot \mathrm{Ent}(\mu \| \mu^*) \quad \text{(log-Sobolev class)}$$

combined with an entropy descent argument could convert the subsequential limit to a full limit, at the cost of restricting to log-Sobolev policy classes. This route remains open.

7.2.4 McKean–Vlasov Games with Asymmetric Information

All agents in the current model observe the same common noise W^0. Extending to *asymmetric information* — where agent i observes a private signal $Y_t^i = h(X_t^i, W_t^0) + \varepsilon_t^i$ in addition to a common signal — leads to a filtering problem for each agent. The equilibrium measure μ_t then becomes the conditional distribution given the agent's information set, and the master equation acquires a Zakai-type SPDE term. This extension connects the MFG framework to nonlinear filtering theory.

7.3 Numerical and Computational Extensions

7.3.1 Deep Galerkin Methods

The finite-difference scheme of Chapter 4 faces a *curse of dimensionality* for $d \geq 2$: the complexity scales exponentially with d. Deep Galerkin methods (DGM, (?)) parametrise the value function $U(t, x, \mu)$ as a neural network and minimise the residual of the master equation (1.7). The challenge is representing the measure argument μ as an input; moment embeddings (using $(\bar{\mu}, v, \ldots)$) or the Sinkhorn attention mechanism can serve as finite-dimensional surrogates. Error analysis for DGM in the MFG setting is an active research area.

7.3.2 Physics-Informed Neural Networks for the Fokker–Planck SPDE

The Fokker–Planck SPDE

$$d\rho_t = -\nabla_x \cdot (\rho_t b_t)\, dt + \frac{\sigma^2}{2} \Delta_x \rho_t\, dt + \sigma_0 \nabla_x \cdot (\rho_t \circ dW_t^0)$$

is a stochastic PDE in $(d+1)$ dimensions. Physics-informed neural networks (PINNs) can enforce the PDE residual in the loss function. For the common-noise case, the stochastic term requires an Itô–Stratonovich correction; implementing this in an automatic-differentiation framework (JAX, PyTorch) is straightforward but has not been systematically studied.

7.3.3 High-Dimensional Common-Noise MFGs

The particle method of Chapter 5 achieves $O(N^{-2/d})$ rate for $d \geq 5$, degrading rapidly with dimension. Three approaches are known to mitigate this: (a) *structured state spaces* where the interaction is through low-dimensional statistics (e.g., first two moments); (b) *normalising flows* that map the measure to a parametric family; (c) *diffusion models* (score-based generative models) that directly approximate μ_t^* from samples. None of these has a rigorous convergence theory in the common-noise MFG setting.

7.3.4 Matrix Riccati Re-derivation for Chapter 8's Multi-Dimensional LQ-MFG

The corrected Appendix A derivation covers the scalar LQ-MFG (Chapter 7 of the original thesis). Chapter 8's alpha-generation model uses a matrix-valued Riccati system, where the same sign corrections likely apply but have not been independently re-derived. Providing a verified matrix re-derivation analogous to Appendix A — including a symbolic PDE-residual check and numerical solver cross-validation — is the primary technical debt left by the revision process and is identified as the first priority for future work.

7.4 Formal Verification Extensions

7.4.1 Full FBSDE Well-Posedness in Lean 4

The current Lean 4 proofs (Chapter 6) cover the key analytic lemmas but not the full FBSDE well-posedness theorem. A complete formalisation would require:

1. Measure-valued SDEs in Lean / Mathlib (not yet available as of Mathlib 4.14; relevant PRs are in progress);
2. The Banach fixed-point theorem applied to the function space $\mathcal{C}([0,T];\mathcal{P}_2(\mathbb{R}^d))$ (available via `Mathlib.Topology.MetricSpace.Contracting`);
3. Measurability of the conditional measure flow $t \mapsto \mu_t$ with respect to $\mathcal{F}_t^{W^0}$ (requires stochastic analysis infrastructure in Lean not yet in Mathlib).

7.4.2 Verified Numerical Scheme

A longer-term goal is a Lean 4 / Coq proof of the convergence theorem (Theorem 1.3) for the Euler–Maruyama scheme. This requires a verified floating-point arithmetic model and interval arithmetic to bound rounding errors. The Lean 4 formalisation of Euler's method for ODEs in `Mathlib.Analysis.ODE.Picard` provides a starting point, but extension to stochastic integrals and Wasserstein distances requires substantial new infrastructure.

7.5 Conclusion

The framework developed in this thesis — quantitative well-posedness (Chapter 3), convergent numerical schemes (Chapter 4), propagation-of-chaos analysis (Chapter 5), and partial formal verification (Chapter 6) — provides a rigorous foundation for the mathematical study of interacting particle systems under common noise. The extensions described in this chapter map out a research agenda that spans stochastic analysis, computational mathematics, and formal methods: exactly the intersection where the most interesting open problems currently lie.

Appendix A

Proofs

This appendix collects the detailed derivations referenced throughout the main text: the algebraic manipulations underlying the Gronwall closure in Theorem 3.9 (Chapter 3), the discretisation error lemma used in Theorem 4.8 (Chapter 4), and the Riccati system derivations underlying the closed-form solutions of Chapters 7 and 8.

A.1 Detailed Derivation of the Contraction Estimate (Chapter 3)

This section provides the algebraic detail omitted from Section 3.5 of Chapter 3, specifically the derivation of the control-difference Lipschitz constant C_α, the BSDE estimate constant C_{BSDE}, and the combination step yielding the differential inequality (3.14).

A.1.1 Derivation of C_α

Recall the optimality condition $\partial_\alpha H(t, X_t, \mu_t, p_t, \alpha_t^*) = 0$. Differentiating implicitly with respect to (x, p, μ) and using the uniform concavity Assumption 3.5 ($\partial_{\alpha\alpha} H \le -\lambda_H < 0$), the implicit function theorem gives

$$\nabla_{(x,p,\mu)} \alpha^* = -\big(\partial_{\alpha\alpha} H\big)^{-1} \nabla_{(x,p,\mu)} \partial_\alpha H.$$

Since $\|\partial_{\alpha\alpha} H\|^{-1} \le 1/\lambda_H$ and $\|\nabla_{(x,p,\mu)} \partial_\alpha H\| \le L_b + L_\mu$ (by the Lipschitz properties of b and the bound on the Lions derivative of f), we obtain

$$C_\alpha = \frac{L_b + L_\mu}{\lambda_H}.$$

This is the constant appearing in (3.10).

A.1.2 Derivation of C_{BSDE}

The linear BSDE (3.11) for Δp_t has driver coefficients A_t, B_t, D_t bounded by the second derivatives of H, which are in turn bounded by Assumption 3.3 (regularity). Standard linear BSDE estimates (El Karoui, Peng, and Quenez, 1997, Proposition 2.1) give

$$\mathbb{E}\Big[\sup_t |\Delta p_t|^2\Big] \le C\,\mathbb{E}\Big[|\Delta X_T|^2 + \int_0^T \big(|\Delta X_t|^2 + |\alpha_t^1 - \alpha_t^2|^2 + W_2(\mu_t^1, \mu_t^2)^2\big)\, dt\Big],$$

where C depends on T, the bounds on A_t, B_t, D_t, and — crucially — on $\|\sigma_0\|_F^2$ through the martingale representation term Δr_t. The common-noise contribution enters because the quadratic variation of the common-noise martingale term in the BSDE energy identity is $\mathbb{E}[\int_0^T |\Delta r_t|^2\, dt]$, and the martingale representation theorem bounds this by the same right-hand side scaled by $\|\sigma_0\|_F^2$. This gives $C_{\text{BSDE}} = C_0'(1 + \|\sigma_0\|_F^2)$ for a constant C_0' depending only on T and the Lipschitz constants.

A.1.3 Combining the Estimates

Substituting (3.10) into (3.9) and (3.13), and writing $y_t = \mathbb{E}[|\Delta X_t|^2] + \mathbb{E}[|\Delta p_t|^2]$, the two coupled inequalities

$$\begin{aligned}
\frac{d}{dt}\mathbb{E}[|\Delta X_t|^2] &\le (2L_b + L_b^2)\,\mathbb{E}[|\Delta X_t|^2] + L_b\,\mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2] \\
&\quad + L_b C_\alpha^2\big(\mathbb{E}[|\Delta X_t|^2] + \mathbb{E}[|\Delta p_t|^2] + \mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2]\big), \\
\mathbb{E}[|\Delta p_t|^2] &\le C_{\text{BSDE}}\,\mathbb{E}\Big[|\Delta X_T|^2 + \int_0^T \big(y_s + \mathbb{E}[W_2(\mu_s^1, \mu_s^2)^2]\big)\, ds\Big]
\end{aligned}$$

can be combined (using Gronwall's inequality on the second to bound $\mathbb{E}[|\Delta p_t|^2]$ pointwise in terms of an integral of y_s, then substituting into the first) into the single differential inequality

$$\frac{d}{dt} y_t \le C_1\, y_t + C_2\,\mathbb{E}[W_2(\mu_t^1, \mu_t^2)^2],$$

with $C_1 = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}$ and $C_2 = L_b + C_\alpha L_b + C_{\text{BSDE}} L_\mu$, where C_{reg} collects the remaining Lipschitz/regularity terms from C_α and C_{BSDE} not already accounted for in the leading $2L_b^2$ and $2\|\sigma_0\|_F^2$ terms. This is precisely (3.14).

A.2 Discretisation Error Lemma (Chapter 4)

This section proves the discretisation error bound used in the proof of Theorem 4.8.

Lemma A.1 (Discretisation error of the fixed-point map)**.** *Let Φ be the continuous-time MFG fixed-point map (Chapter 3) and Φ^Δ its discrete analogue (Chapter 4, Section 4.2). Under Assumptions 1.13–1.19 and the smoothness Assumption 4.1, for any measure flow $\mu \in \mathcal{M}$,*

$$\|\Phi^\Delta(\mu) - \Phi(\mu)\|_{\lambda,T} \le C_{\text{disc}}(\Delta t^{1/2} + \Delta x),$$

where C_{disc} depends on the Lipschitz/smoothness constants but not on μ, Δt, or Δx.

Proof. Fix $\mu \in \mathcal{M}$. By definition, $\Phi(\mu)$ is obtained by solving the continuous-time agent FBSDE (3.5) exactly, while $\Phi^\Delta(\mu)$ is obtained from the discrete scheme (4.1)–(4.5). We bound the difference in three steps, corresponding to the three discretised components.

Step 1 (forward SDE). By Lemma 4.3, the weak error between the continuous state X_t and discrete state X^n (at matching times $t_n = n\Delta t$) is $O(\Delta t)$ for smooth test functions. Combined with Lemma 4.5, this gives $\mathbb{E}[|X^n - X_{t_n}|^2] \le C\Delta t$, hence a strong-error contribution of order $\Delta t^{1/2}$ to the Wasserstein distance between laws.

Step 2 (adjoint BSDE). By Lemma 4.4, the discrete adjoint process satisfies $|\mathbb{E}[\psi(p^n)] - \mathbb{E}[\psi(p_{t_n})]| \le C\Delta t$ for smooth ψ. This propagates to the optimal control via the Lipschitz dependence $\alpha^* = p/\lambda$ (**??**), contributing a further $O(\Delta t)$ term, which after the same square-root argument as Step 1 contributes $O(\Delta t^{1/2})$ to the Wasserstein bound.

Step 3 (Fokker–Planck SPDE / measure consistency). By Theorem 4.6, the finite-difference (or particle) discretisation of the measure flow satisfies $\sup_n \mathbb{E}[\|\rho^n - \rho_{t_n}\|_{L^2}^2]^{1/2} \le C(\Delta t^{1/2} + \Delta x)$. Since the L^2 density norm controls the Wasserstein-2 distance for measures with bounded density ratios (a consequence of the Cauchy–Schwarz/transport-cost duality; see Villani, 2009, Remark 6.6), this contributes the dominant $O(\Delta t^{1/2} + \Delta x)$ term.

Summing the three contributions (each bounded uniformly over μ by the Lipschitz continuity of the coefficients, which ensures the discretisation constants do not depend on which measure flow μ is plugged into Φ versus Φ^Δ) gives

$$\|\Phi^\Delta(\mu) - \Phi(\mu)\|_{\lambda,T} \le C_{\text{disc}}(\Delta t^{1/2} + \Delta x),$$

as claimed, with C_{disc} depending only on the Lipschitz and smoothness constants of b, f, g and the CFL-condition constants, uniformly over bounded sets of measure flows. □

A.3 Derivation of the Corrected LQ-MFG Riccati System

This section provides the detailed derivation of the Riccati equations for the canonical linear-quadratic MFG (Example 1.6). The scalar case is derived in full here; the matrix generalisation for multi-dimensional state spaces follows by an identical argument with matrix-valued coefficients and is flagged in Chapter 7 as remaining future work.

A.3.1 Setup

We work with the scalar LQ-MFG (Example 1.6). The matrix generalisation follows by an identical coefficient-matching argument with matrix products replacing scalar ones. Substituting the quadratic ansatz (**??**) into the HJB equation

$$\partial_t u + \frac{\sigma^2}{2}\partial_{xx} u + \frac{\sigma_0^2}{2}\partial_{xx} u + H^*(t, x, \mu, \partial_x u) + \int_{\mathbb{R}} \frac{\delta u}{\delta \mu}(\mu)(y) \cdot b(t, y, \mu, \alpha^*)\, \mu(dy) = 0,$$

and matching coefficients of x^2, $x\bar{\mu}$, $\bar{\mu}^2$, x, $\bar{\mu}$, and constants yields the system of ODEs. The Lions derivative of u with respect to μ at y is

$$\frac{\delta u}{\delta \mu}(\mu)(y) = -B_t x - C_t \bar{\mu} - E_t.$$

Substituting into the integral term and using $\int y\, \mu(dy) = \bar{\mu}$ gives the contributions to the coefficients. The terms involving D_t, E_t, F_t arise from the linear part of the Hamiltonian and the bailout policy b_t (Chapter 7) or the baseline drift (Chapter 8, where $d_t = e_t = f_t = 0$ since Example 1.1 has no exogenous linear forcing term).

A.3.2 Quadratic Terms

Correction (this revision): the derivation below replaces an earlier version that, despite an empirical optimality check confirming the *sign* of the $A_t B_t/\lambda$ cross term in $\dot{B}_t$, was found on full re-derivation to (i) contain an independent sign error in $\dot{A}_t$ unrelated to that check, and (ii) omit a Lions-derivative (master-equation) coupling term from $\dot{B}_t$, $\dot{D}_t$, $\dot{C}_t$, $\dot{E}_t$, and $\dot{F}_t$ alike. Both issues are corrected together here, since they are intertwined: the corrected A_t feeds into every other coefficient. The correction was found by (a) re-deriving the Hamiltonian directly from the cost functional and adjoint BSDE of Section **??** rather than from the previously stated H^* alone, which pins down the maximised Hamiltonian and its terminal condition unambiguously, and (b) deriving the mean-field coupling term explicitly from the same conditional-mean consistency argument used for the aggregate drift $\bar{b}(t, \bar{\mu})$ below. The full corrected system was verified three independent ways: (1) direct symbolic substitution showing the PDE residual vanishes identically; (2) numerical integration with two independent solvers (LSODA and Radau) agreeing to all displayed digits; (3) the corrected A_t, B_t, D_t equations, when stripped of the mean-field coupling term, exactly reproduce the maximised-Hamiltonian-only matching below, confirming internal consistency of the two-step derivation.

The cost functional (Eq. **??** and surrounding text) and adjoint BSDE (Eq. **??**) give the Hamiltonian

$$H(t, x, \mu, p, \alpha) = p\big[\kappa(\bar{\mu} - x) + \alpha + b_t\big] - \frac{c_1}{2}x^2 - \frac{\lambda}{2}\alpha^2,$$

maximised over α (the bank maximises negative cost) at $\alpha^* = p/\lambda$, giving the maximised Hamiltonian

$$H^*(t, x, \mu, p) = p \cdot \kappa(\bar{\mu} - x) + \frac{p^2}{2\lambda} + pb_t - \frac{c_1}{2}x^2,$$

identical in form to the previously stated H^*. With $p = \partial_x V = -A_t x - B_t\bar{\mu} - D_t$ (confirmed via the adjoint terminal condition $p_T = -c_2 X_T$, which matches $\partial_x V(T, x, \mu) = -c_2 x$ exactly when $A_T = c_2, B_T = D_T = 0$), matching coefficients of x^2 in $\partial_t V + H^* = 0$ gives

$$\dot{A}_t = 2\kappa A_t + \frac{A_t^2}{\lambda} - c_1, \qquad A_T = c_2.$$

This differs from the previously stated equation in the sign of both the A_t^2/λ and c_1 terms. The corrected equation is the standard sign for a cost-*minimisation* Riccati equation (equivalently, a value-*maximisation* equation written with $p = \partial_x V$), and was independently verified by formulating the bank's problem as a textbook two-state $(x, \bar{\mu})$ linear-quadratic regulator and solving the resulting matrix Riccati equation directly, which reproduces this same A_t with no free sign choices. A welcome consequence: under this corrected equation, the numerical illustration of Section **??** no longer exhibits the finite-time blow-up reported in an earlier revision, and the originally intended parameter $c_2 = 5.0$ may be restored without modification (see the updated Table **??** and discussion in Section **??**).

The full master equation for $V(t, x, \mu) = -\frac{1}{2}A_t x^2 - B_t x\bar{\mu} - \frac{1}{2}C_t\bar{\mu}^2 - D_t x - E_t\bar{\mu} - F_t$ also requires the Lions-derivative coupling term, which (since $\delta V/\delta\mu(y) = -B_t x - C_t\bar{\mu} - E_t$ is independent of y) reduces to $\bar{b}(t, \bar{\mu}) \cdot \partial_{\bar{\mu}} V$, where $\bar{b}(t, \bar{\mu}) = \mathbb{E}_{y\sim\mu}[\kappa(\bar{\mu} - y) + \alpha^*(t, y, \mu) + b_t]$ is the equilibrium aggregate drift of $\bar{\mu}_t$ itself. Since $\alpha^*(t, y, \mu) = p(t, y, \mu)/\lambda$ is linear in y, this expectation needs only the first moment $\mathbb{E}[y] = \bar{\mu}$ (no higher-moment terms enter), giving exactly

$$\bar{b}(t, \bar{\mu}) = -\frac{(A_t + B_t)\bar{\mu} + D_t}{\lambda} + b_t.$$

Together with the common-noise diffusion terms $\frac{1}{2}(\sigma^2 + \sigma_0^2)\partial_{xx}^2 V + \sigma_0^2\partial_{x\bar{\mu}}^2 V + \frac{1}{2}\sigma_0^2\partial_{\bar{\mu}\bar{\mu}}^2 V$ – present because $\bar{\mu}_t$ is itself a stochastic process driven by the same common noise σ_0 as

X_t – the full master-equation HJB is

$$\partial_t V + H^*(t, x, \mu, \partial_x V) + \bar{b}(t, \bar{\mu})\partial_{\bar{\mu}} V + \tfrac{1}{2}(\sigma^2 + \sigma_0^2)\partial^2_{xx} V + \sigma_0^2 \partial^2_{x\bar{\mu}} V + \tfrac{1}{2}\sigma_0^2 \partial^2_{\bar{\mu}\bar{\mu}} V = 0.$$

The coupling term $\bar{b} \cdot \partial_{\bar{\mu}} V$ contributes *only* to the $x\bar{\mu}$, $\bar{\mu}^2$, x, $\bar{\mu}$, and constant coefficients (verified symbolically: its contribution to the x^2 coefficient is identically zero, so A_t above is unaffected by this correction). Matching coefficients of $x\bar{\mu}$ now gives

$$\dot{B}_t = \kappa B_t - \kappa A_t + \frac{2A_t B_t}{\lambda} + \frac{B_t^2}{\lambda}, \qquad B_T = 0,$$

which agrees with the previously corrected sign of the $A_t B_t/\lambda$ term (confirming that empirical check remains valid) but adds the $2A_t B_t/\lambda + B_t^2/\lambda$ master-equation contribution, replacing the single $A_t B_t/\lambda$ term that appeared before. Matching coefficients of $\bar{\mu}^2$:

$$\dot{C}_t = -2\kappa B_t + \frac{2A_t C_t}{\lambda} + \frac{B_t^2}{\lambda} + \frac{2B_t C_t}{\lambda}, \qquad C_T = 0.$$

These replace (**??**) (and, with Q, P in place of the scalar $c_1/2, c_2/2$ and matrix products throughout, the analogous system in Chapter 8's (**??**) should be re-derived under the same correction; this is flagged as remaining work in Chapter 10, §10.4, since the matrix case was not re-derived in this revision).

A.3.3 Linear Terms

Matching coefficients of x:

$$\dot{D}_t = \kappa D_t + \frac{A_t D_t}{\lambda} + \frac{B_t D_t}{\lambda} - (A_t + B_t)b_t, \qquad D_T = 0,$$

which adds the $B_t D_t/\lambda - B_t b_t$ master-equation contribution to the previously stated equation. Matching coefficients of $\bar{\mu}$:

$$\dot{E}_t = -\kappa D_t + \frac{A_t E_t}{\lambda} + \frac{B_t D_t}{\lambda} + \frac{B_t E_t}{\lambda} + \frac{C_t D_t}{\lambda} - (B_t + C_t)b_t, \qquad E_T = 0.$$

This confirms the open item flagged in an earlier revision: $\dot{E}_t$ does indeed depend on C_t, through the $C_t D_t/\lambda$ term. Matching the constant term, which collects the diffusion contributions $\partial^2_{xx} V = -A_t$, $\partial^2_{x\bar{\mu}} V = -B_t$, $\partial^2_{\bar{\mu}\bar{\mu}} V = -C_t$ together with the remaining $\bar{b} \cdot \partial_{\bar{\mu}} V$ pieces:

$$\dot{F}_t = \frac{D_t^2}{2\lambda} + \frac{D_t E_t}{\lambda} - (D_t + E_t)b_t - \frac{\sigma^2 + \sigma_0^2}{2} A_t - \sigma_0^2 B_t - \frac{\sigma_0^2}{2} C_t, \qquad F_T = 0.$$

The full system $(A_t, B_t, C_t, D_t, E_t, F_t)$ was verified to make the PDE residual vanish identically under direct symbolic substitution, and was integrated numerically (Section **??**) with no blow-up across the full original parameter range, including the originally intended $c_2 = 5.0$. As in the previous revision, when $b_t \equiv 0$ (Chapter 8's alpha-generation model has no exogenous bailout term), $D_t = E_t \equiv 0$, but C_t and F_t no longer vanish in general – the simplification used in Chapter 8 should therefore be re-checked against this corrected system as part of the matrix-case re-derivation noted above.

A.3.4 Proof that v_t is Independent of b_t

From the variance dynamics (**??**), the variance v_t satisfies a linear ODE whose coefficients depend only on $A_t, \kappa, \lambda, \sigma$, and σ_0. None of these depend on the bailout policy b_t. Therefore, the government's choice of b_t affects the mean $\bar{\mu}_t$ but not the variance v_t. This justifies the separation in the government's objective used to derive (**??**).

A.3.5 Government's Riccati Equation (Chapter 7)

The government's value function takes the form $V(t, \bar{\mu}) = -\frac{1}{2}\Pi_t \bar{\mu}^2 - \delta_t$. Substituting into the HJB equation for the government's control problem (with dynamics (**??**) and cost (**??**)) and matching coefficients of $\bar{\mu}^2$ yields (**??**).

A.3.6 Explicit Bound for the Propagation of Chaos Constant (Chapter 7)

The bound for C in Proposition **??** follows from the synchronous coupling estimate of Chapter 3. Specifically, let $\Delta X_t^i = X_t^{i,N} - X_t^{i,*}$ be the difference between the N-bank state and the limiting McKean–Vlasov state under the same noises. Then, by the same Itô-expansion argument as in Appendix A.1,

$$\mathbb{E}\big[|\Delta X_t^i|^2\big] \leq \int_0^t e^{2L_b(t-s)} L_b \, \mathbb{E}\big[W_2(\mu_s^N, \mu_s^*)^2\big] \, ds.$$

Summing over i and using $\mathbb{E}[W_2(\mu_t^N, \mu_t^*)^2] \leq \frac{1}{N}\sum_{i=1}^N \mathbb{E}[|\Delta X_t^i|^2]$ yields the claimed bound via Gronwall's inequality, with the explicit constant

$$C \leq \frac{4}{\kappa_{\min}} \left(\sigma^2 + \sigma_0^2 + \frac{1}{\lambda^2} \sup_t (A_t + B_t)^2 \right) e^{2L_b T},$$

where $\kappa_{\min} = \inf_t(\kappa + A_t/\lambda) > 0$ and $L_b = \kappa + \sup_t(A_t + |B_t|)/\lambda$, as stated in Chapter 7.

Appendix B

Additional Results

This appendix consolidates two pieces of reference material that are useful when reading the thesis non-linearly: a complete index of every numbered theorem, proposition, and lemma across all ten chapters (Appendix B.1), and an expanded notation glossary that supplements the brief table in Section 1.9 (Appendix B.2).

B.1 Index of Numbered Results

Tables B.1 to B.3 list every theorem, proposition, and named lemma in the thesis, in order of appearance, together with the chapter in which it is proved and a one-line statement for quick lookup.

Reading guide. The three results in **bold** (Theorems 3.1, 4.1, 5.1, 5.2, and Propositions 6.1, 8.1) form the central logical chain summarised in the conclusion of Chapter 10: well-posedness → numerical approximability → learnability → practical architecture → tradable signals.

B.2 Expanded Notation Glossary

Table B.4 expands on the brief notation table of Section 1.9.1, adding symbols introduced in later chapters (the systemic-risk and alpha-signal notation of Chapters 7–9 in particular).

Remark B.1 (Notational overload). A small number of symbols are deliberately reused across chapters with related but distinct meanings, mirroring the source material: κ_t denotes the mean-reversion *signal* in Chapter 8 but the effective mean-reversion *rate* $\kappa + A_t/\lambda$ wherever it appears in Chapters 3–7; γ denotes a bailout cost in Chapter 7 but a risk-aversion coefficient in Chapter 8; and the RL learning-rate sequence γ_t (Chapters 5–6) is unrelated to the portfolio risk-aversion parameter γ (Chapter 8). Context disambiguates in every case, but readers jumping directly to a single chapter should consult Table B.4 rather than assuming global consistency.

Table B.1: Index of numbered results: Chapter 2 (mathematical toolkit)

Chapter	**Result**	**One-line statement**
2	Theorem 2.2 (Brenier)	Optimal transport map between absolutely continuous measures is the gradient of a convex function.
2	Theorem 2.5	$(\mathcal{P}_2(\mathbb{R}^d), W_2)$ is a complete separable metric space.
2	Theorem 2.6	W_2-convergence $\iff$ weak convergence + second-moment convergence.
2	Theorem 5.3 (Fournier–Guillin)	Empirical measure converges to μ in W_2 at rate $N^{-1/2}$ $(d = 1)$ or $N^{-1/d}$ $(d \geq 3)$.
2	Theorem 2.19 (Itô isometry)	$\mathbb{E}[(\int H\, dB)^2] = \mathbb{E}[\int H^2\, dt]$.
2	Theorem 2.22 (Itô's formula)	Multidimensional chain rule for Itô processes with idiosyncratic and common noise.
2	Theorem 2.26	Lipschitz + linear growth $\Rightarrow$ unique strong solution to the SDE.
2	Theorem 2.28	The conditional law of a McKean–Vlasov SDE solves the Fokker–Planck SPDE.
2	Theorem 2.31	The Fokker–Planck SPDE is well-posed in $L^2(\Omega; H^1)$.

Table B.2: Index of numbered results: Chapter 2 (continued)

Chapter	**Result**	**One-line statement**
2	Theorem 2.35 (Doob)	$\mathbb{E}[\sup_t M_t^2] \leq 4\mathbb{E}[M_T^2]$ for non-negative submartingales.
2	Theorem 2.37 (Burkholder–Davis–Gundy)	$\mathbb{E}[\sup_t \lvert M_t\rvert^p] \sim \mathbb{E}[[M,M]_T^{p/2}]$.
2	Theorem 2.38	Every martingale in a Brownian filtration is an Itô integral.
2	Theorem 2.40 (Girsanov)	Change-of-measure formula removing a drift via an exponential martingale.
2	Theorem 2.42	$H^k \hookrightarrow C^0$ for $k > d/2$.
2	Theorem 2.43 (Hahn–Banach)	Norm-preserving extension of a bounded linear functional.
2	Theorem 2.46 (Lumer–Phillips)	Dissipative + dense range $\Rightarrow$ generates a contraction semigroup.
2	Theorem 2.49	Log-Sobolev inequality implies a spectral gap lower bound.
2	Theorem 2.54 (Stochastic Maximum Principle)	Optimal control maximises the Hamiltonian pointwise.
2	Theorem 2.55	SMP adjoint variables equal derivatives of the HJB value function.

Table B.3: Index of numbered results: Chapters 3–8 (main theoretical and applied contributions)

Chapter	Result	One-line statement
3	**Theorem 3.9** (Quantitative Well-Posedness)	Unique MFG equilibrium exists; Φ is a contraction in weighted W_2 with explicit constant $C = 2L_b^2 + 2\|\sigma_0\|_F^2 + C_{\text{reg}}$.
4	**Theorem 4.8** (Numerical Convergence)	Combined Euler–Maruyama/particle/SPDE scheme converges with weak error $O(\Delta t^{1/2} + \Delta x)$.
4	Theorem 4.6	Finite-difference Fokker–Planck scheme converges at the same rate under a CFL condition.
5	**Theorem ??** (Subsequential RL Convergence)	Three-timescale actor-critic converges $\liminf W_2 = 0$ a.s. under compact Lipschitz function classes.
5	**Theorem ??** (Exponential Convergence)	Entropy regularisation + log-Sobolev policy class $\Rightarrow$ exponential W_2 convergence rate $\lambda = \beta/(2C_{\text{Lip}})$.
6	**Proposition ??** (Hierarchical Architectural Principle)	SPDE operator decomposition justifies a fast-critic/intermediate-actor/slow-meta-learner timescale separation.
7	Proposition **??**	$O(1/N)$ propagation of chaos and $\varepsilon_N = O(N^{-1/2})$-Nash property for the systemic risk model.
8	**Proposition ??** (Four Alpha Signals)	Optimal spread, mean-reversion speed, factor exposure, and crowding measure $c_t = W_2(\mu_t, \mu_t^*)$ are explicit functions of the LQ-MFG equilibrium.

Table B.4: Expanded notation glossary

Symbol	**Meaning**
X_t, X_t^i	State (log-position / bank reserve) of the representative or i-th agent
μ_t, μ_t^N	Population measure flow; empirical measure of N agents
μ_t^*	MFG equilibrium measure flow
$\bar{\mu}_t, \bar{x}_t$	Population mean (first moment of μ_t)
α_t, α_t^*	Control (trading rate / capital-raising effort); optimal control
b_t	Government bailout rate (Chapter 7 only)
p_t, q_t, r_t	Adjoint (costate) process and its idiosyncratic/common-noise volatility components
κ	Mean-reversion speed
λ	Cost-of-control (trading cost / effort cost) parameter
γ	Cost-of-bailout parameter (Ch. 7) or risk-aversion parameter (Ch. 8)
σ, σ_0	Idiosyncratic and common-noise volatility
A_t, B_t, C_t	Quadratic Riccati coefficients (value-function ansatz)
D_t, E_t, F_t	Linear/constant Riccati coefficients (Ch. 7–8)
Γ_t, Π_t	Effective mean-reversion rate and government Riccati coefficient (Ch. 7)
c_t	Crowding measure, $c_t = W_2(\mu_t, \mu_t^*)$
$s_{i,t}$	Optimal spread signal for asset i
κ_t	(Overloaded in Ch. 8) model-implied mean-reversion speed signal, $\kappa + A_t/\lambda$
$\beta_{i,t}$	Factor exposure signal for asset i
$\delta(c_t)$	Crowding discount function applied to portfolio weights
w_t^{MV}, w_t^*	Baseline mean-variance weights; crowding-discounted final weights
θ	Generic vector of LQ-MFG model parameters to be calibrated
$\hat{\theta}$	GMM estimator of θ
$\gamma_t, \beta_t, \eta_t$	(Ch. 5–6, overloaded with the risk-aversion γ of Ch. 8) fast/intermediate/slow RL learning-rate sequences
θ_t, ϕ_t, ψ_t	Critic, actor, and meta-learner parameters (Ch. 5–6)
$\mathcal{L}_{\mathrm{idio}}$, $\mathcal{L}_{\mathrm{mf}}$, $\mathcal{L}_{\mathrm{common}}$	SPDE operators for idiosyncratic diffusion, mean-field interaction, and common-noise transport (Ch. 6)

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